
Time-Conditioned Foreseeing: An EHR-Specific Foundation Model for Irregular Dynamics and Calendrical Time

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Abstract

Electronic Health Records (EHRs) possess unique characteristics distinct from natural language, yet existing EHR foundation models often rely on suboptimal NLP-based approaches. We propose a pretraining method tailored to EHRs’ distinct features. First, we introduce Pathology-Focused Binning, a density-based quantization strategy that prioritizes clinically significant numerical ranges over usual values. Second, to jointly capture both the exact timing of clinical events and the relative intervals between them, we propose Dual-Calendar Rotary Positional Embedding (RoPE), which encodes absolute and relative temporal signals. Third, we introduce the Time-Conditioned Foreseeing (TCF) objective, aligning with clinical treatment planning to forecast events across multiple temporal horizons by explicitly modeling event timing. Our approach establishes a temporal generative EHR model that outperforms existing foundation models on eleven diverse downstream tasks—achieving up to a 54% improvement in AUPRC—and enables the generation of realistic, temporally consistent patient trajectories. Code is available at https://github.com/Pusheen-cat/TCF_PFM.

1. Introduction

Electronic health records (EHRs) are longitudinal data that comprehensively document a patient’s medical history. These records enable clinicians to assess patient status, coordinate interventions, and facilitate communication within healthcare teams (Häyrinen et al., 2008). Accordingly, one

of the key objectives in medical AI is to develop models capable of performing diverse clinical tasks by learning from EHRs. Recently, significant efforts have been made to adapt training paradigms from large language models (LLMs) to the EHR domain (Niu et al., 2024). However, a fundamental gap remains in addressing the distinct properties of EHR data that differentiate it from natural language. Consequently, recent architectures struggle to effectively model numerical values, capture temporal dynamics, and establish learning objectives that reflect the complexities of clinical decision-making (Nasarudin et al., 2024).

EHRs comprise diverse clinical events—such as examinations, treatments, and diagnoses—each associated with a timestamp. Recent studies commonly represent each event (E) as a Time (T), Feature (F), Value (V) triplet (Tipirneni & Reddy, 2022; Lee et al., 2023) as illustrated in Figure 1. In this framework, F denotes elements like diagnosis codes or laboratory tests (e.g., Systolic Blood Pressure), while V represents numerical results or auxiliary information (e.g., 87 mmHg). Although a faithful representation of clinical trajectories requires all three components, certain recent foundation models still exclude Time or Value due to modeling complexities (Yang et al., 2024), as indicated in the “data usage” column of Table 1.

Recent advances in LLMs have catalyzed the development of EHR foundation models that leverage large-scale pre-training (Li et al., 2020). While many current models now incorporate Time-Feature-Value triplet, their underlying architectures and learning objectives often remain unaligned with the clinical semantics of EHR data, leading to suboptimal performance (Burkhart et al., 2025). Specifically, we identify three critical limitations in existing paradigms: (1) Inadequate numerical representation, where quantile binning fails to prioritize clinically significant pathological values; (2) Temporal signal loss, where standard positional embeddings overlook calendrical context or relative intervals (Likhomanenko et al., 2021); and (3) Short-sighted learning objectives, where NLP pre-training objectives like next-token prediction (NTP) or masked language modeling (MLM) fall short on the long-term clinical planning. To address these limitations, we introduce improved binning, temporal embedding, and novel training objectives

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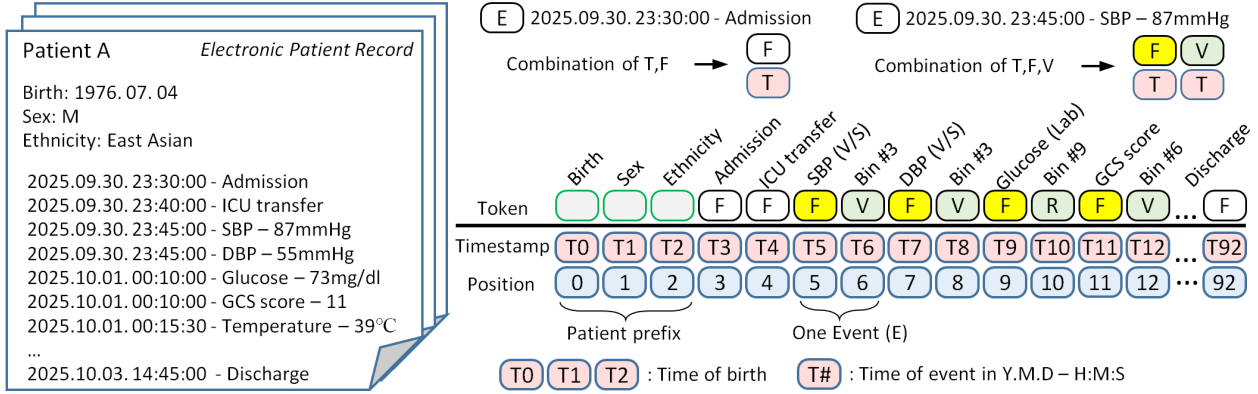


Figure 1. (Left) Raw patient data from the EHR database in chronological order. (Right) Tokenization of each event (E) with triplet representation. Patient information is placed at the beginning. Features and Values are tokenized, and timestamps remain continuous.

specifically tailored to the distinct structure of EHR data.

First, we introduce **Pathology-focused Binning**, a density-based quantization strategy for Value tokenization. As shown in Table 1 [Value Binning], most EHR models rely on quantile binning, which disproportionately assigns bins to dense physiologic ranges while providing insufficient resolution for sparse, clinically critical pathologic ranges (Figure 2A). Other approaches rely on false distributional assumptions of Gaussianity—such as standard deviation-based binning (Zhu et al., 2024) or z-normalization (Tipirneni & Reddy, 2022)—making them vulnerable to the outliers and multi-modal distributions common in EHR data. To address these limitations, our approach makes no distributional assumptions. Instead, it assigns lower weights to high-density physiologic zones and higher weights to low-density pathologic zones. This design ensures sufficient resolution for abnormal values across diverse distributions, representing the first application of such density-based binning to EHR foundation models.

Second, we introduce **Dual-Calendar RoPE**, a novel timestamp addressing method for EHRs. Unlike natural language, where tokens are uniformly spaced, EHR events occur at highly irregular intervals. Clinically, capturing both relative intervals and calendrical context—such as diurnal variations in body temperature or the increased risk of dialysis complications following weekends (Fotheringham et al., 2020)—is essential. Furthermore, EHRs often contain concurrent events, such as laboratory tests recorded together. As shown in Figure 2B, we partition the dimensions of rotary positional embedding (Su et al., 2024) to jointly encode position and time, assigning calendrical components (e.g., minute, day, month) in increasing units to the time dimension. This enables explicit modeling of distance relations such as “two tests performed simultaneously”¹ or “identical

¹For simultaneous SBP, 120mmHg, DBP, 80mmHg record, the position dimension provides support to prevent the model from confusing results such as SBP, 80mmHg, DBP, 120mmHg.

tests performed at the same hour on different days.” As indicated in Table 1 [Time Addressing], conventional models lack such multi-faceted temporal addressing capabilities.

Finally, we propose **Time-Conditioned Foreseeing (TCF)**, a learning objective that mirrors the clinical treatment planning and enables timestamped EHR generation for the first time. As shown in Table 1 [Learning Objective], prior models have relied on objectives designed for LLMs or variants thereof (time-to-event; TTE objective is an exception). Conventional NTP loss capture only $P(F_{next} | E_{past})$, without explicitly modeling temporal information. Consequently, they cannot distinguish whether an event occurs minutes later or after many hours, treating both urgent and routine vital sign measurements identically as the ‘next token.’

In contrast, TCF explicitly models long-range temporal information, thereby capturing how real-world clinical practice unfolds over time. In NLP, missing a single token disrupts grammar, and consecutive tokens are tightly correlated. By contrast, neighboring EHR events are loosely connected and often exhibit long-range dependencies, such as 8-hour follow-up tests. This reflects clinical practice, where physicians do not always act in real time but instead devise broader clinical plans. TCF embodies this principle: rather than the short-sighted scope of NTP, which predicts only the immediate next event, TCF enables questions such as, “What intervention is needed in the next six hours?” To achieve this, TCF module first generates the next timestamp from the last hidden state. The multiple foreseeing timestamps are then fed back as module inputs, conditioning subsequent token generation. This time-conditioned architecture allows simultaneous learning of $P(T_{next} | E_{past})$ and $P(F_{foresees} | T_{foresees}, E_{past})$, leading to improved performance. Our model ranked first across all three dataset configurations and seven diverse downstream tasks. Across these tasks, the AUPRC was consistently improved, reaching up to 54% higher than the eleven-best model, highlighting a clinically meaningful improvement given the data

Table 1. Comparison of recent EHR models by architecture and training objective. **[Data usage]**: whether the model uses Time and Value information. **[Value binning]**: whether values are discretized using non-quantile binning (STraTS embeds values continuously) and whether bin tokens are shared across Features. **[Time addressing]**: whether the model considers relative time intervals, calendrical time, and uses an order-preserving tie-breaker for concurrent events. **[Learning objective]**: type of loss, ‘Foresee’ - whether model forecasts beyond the next-token, and ‘Temporal Generation’ - whether temporal generative modeling is possible. Abbreviations*: MEP; missing entity prediction, TTE; time to event prediction, TCF; time conditioned foreseeing. Refer to the related works section for details[†].

Models	[Data usage]		[Value Binning]		[Time addressing]			[Learning Objective]		
	Event Timestamp	Numeric Value	Non-quantile	Bin sharing	Relative interval	Calendrical time	Order-preserving	Type	Foresee	Temporal Generation
BEHRT (2020) Med-BERT (2021) Foresight (2024) ClinicalMamba (2024) EHR-BERT (2024)	X	X	-	-	X	X	-	NTP&MLM	X	X
HEART (2024) FM4EHR (2025)	X	O	X	X O	X	X	-	MEP* NTP	X	X
MOTOR (2024)	O	X	-	-	O	X	X	TTE*	O	X
STraTS (2022)	O	O	O	O	X	X	-	MSE	X	X
EHRSHOT (2023)	O	O	X	X	O	X	X	NTP	X	X
TRADE (2024)	O	O	O	X	X	X	-	MLM	X	X
EHRMamba (2025)	O	O	X	X	X	X	-	NTP&MLM	X	X
ETHOS (2024)	O	O	X	O	$\Delta^{\dagger}$	X	-	NTP	X	$\Delta^{\dagger}$
OURS	O	O	O	both	O	O	O	TCF*	O	O

imbalance. We also demonstrated that the model generates temporally stable, realistic EHR records and leverages the calendrical component in generative modeling.

Our contributions can be summarized as follows:

- Pathology-Focused binning: Introduces density-adjusted binning to the EHR foundation model, focusing on clinically relevant pathologic ranges.
- Dual-Calendar RoPE: Simultaneously represents calendrical/relative time and positional information, capturing calendrical periodicity and avoid concurrency.
- Time Conditioned Foresee Objective: Enables clinically aligned foreseeing training and temporal generative modeling of patient medical timelines.

2. Related works

Table 1 summarizes recent EHR foundation models. These models learn directly from raw EHR events (Burkhart et al., 2025) and support a wide range of downstream clinical tasks. EHR models are distinct from medical specialist LLMs, which rely on clinical texts summarized by clinicians.

A common practice in EHR modeling is to represent each event as (T, F, V) triplets (Lee et al., 2023). However, many models exclude temporal and numeric data, as they are difficult to handle in standard LLM frameworks. For instance, BEHRT (Li et al., 2020), Med-BERT (Rasmy et al., 2021), and others rely solely on discrete Features, omitting critical information and limiting their utility. Foresight (Kral-

jevic et al., 2024) employs learned PE and provides basic temporal addressing with age- and day-level separation.

Some models incorporate numeric Value but omit Time. HEART (Huang et al., 2024) discretizes Values into quantile bins, mapping Feature-Value pairs to single tokens. In contrast, FM4EHR (Burkhart et al., 2025) tokenizes Features and Values separately, sharing bin tokens across features. While feature-wise bin tokens shorten sequences and improve training efficiency, shared bin tokens reduce vocabulary size and data sparsity.

In contrast, MOTOR (Steinberg et al., 2024) models Time but not Value, treating each feature occurrence as a survival analysis endpoint. Its utility is limited by the exclusion of numerics, coarse temporal resolution due to pre-defined intervals, and an unrealistic constant hazard assumption. Moreover, encoding timestamps merely as ‘days since birth’ via RoPE fails to capture calendrical patterns.

STraTS (Tipirneni & Reddy, 2022) tokenizes only the Feature, embedding Value and Time as continuous variables to predict the next value. By modeling only $P(V_{next} | E_{past})$, it loses important context and cannot support generative modeling.

TRADE (Zhu et al., 2024) and EHRMamba (Fallahpour et al., 2025) adopt MLM/NTP paradigms, discretizing values and embedding Feature and Value tokens. EHRMamba applies Time2Vec-based time vectorization, employing absolute positional embeddings (APEs).

ETHOS (Renc et al., 2024) tokenizes time intervals and insert time-interval tokens between events. This coarse

discretization limits medical precision, causes cumulative errors, and increases computational cost by lengthening the sequence. Unlike positional embeddings, it requires aggregating all intervening tokens to determine a time duration.

We provide detailed descriptions of each model and justifications for Table 1 in Appendix A.

3. Method

In this work, we represent a patient’s medical history as a sequence of events $E_{1:L}$, where each event is a triplet $E_i = (T_i, F_i, V_i)$ denoting the timestamp, feature, and value, respectively. When these events are arranged into an input sequence, we denote the discrete token position by P .

Pathology-Focused Binning. To handle numerical values effectively, we employ a non-parametric discretization strategy that prioritizes sparse regions. Let $\mathcal{V}^f = \{v_1, v_2, \dots, v_N\}$ be the multiset of all scalar values observed for a specific feature f in the training set, and let σ_f denote the standard deviation of $\mathcal{V}^f$.

First, we estimate the probability density function of the values. We define a set of uniformly spaced grid points $\mathcal{G} = \{g_1, g_2, \dots, g_M\}$ spanning $[\min(\mathcal{V}^f), \max(\mathcal{V}^f)]$ with a step size of $0.05\sigma_f$. At each grid point $g_k \in \mathcal{G}$, the local density $\hat{p}(g_k)$ is approximated using a Gaussian Kernel Density Estimator:

$$\hat{p}(g_k) = \frac{1}{N} \sum_{j=1}^N K_h(g_k - v_j), \text{ where } K_h(u) = \exp\left(-\frac{u^2}{2h^2}\right)$$

Here, the bandwidth is set to $h = 0.1\sigma_f$. Based on this estimation, we assign weight $w(v)$ to every value v inversely proportional to its local density: $w(v) \propto \hat{p}(v)^{-\lambda}$ (with hyperparameter $\lambda > 0$). This mechanism amplifies the significance of values in sparse, "pathological" regions.

Second, we perform weighted percentile binning. Let $U^f = \{u_1, \dots, u_K\}$ be the set of unique values in $\mathcal{V}^f$. For each unique value u_j , let c_j be its raw count (frequency) in the dataset. We compute a weighted count $\tilde{c}_j$:

$$\tilde{c}_j = c_j \cdot w(u_j)$$

The final bin thresholds are determined by the cumulative distribution of these weighted counts $\{\tilde{c}_j\}_{j=1}^K$. Consequently, high-weight values in pathological ranges occupy a larger portion of the cumulative space, resulting in finer-grained binning in clinically critical areas (Figure 2A). More detailed methodology is described in Appendix B.1.

Dual-Calendar Rotary Position Embedding. We introduce a novel positional encoding tailored for EHR temporal characteristics, modeling both sequential order and cyclical calendrical time (Figure 2). Let $\mathbf{z} \in \mathbb{R}^d$ denote the query or

key vector for an attention head. We partition $\mathbf{z}$ into two sub-vectors: a positional component $\mathbf{z}_{\text{pos}} \in \mathbb{R}^{d_{\text{pos}}}$ and a temporal component $\mathbf{z}_{\text{time}} \in \mathbb{R}^{d_{\text{time}}}$, such that $d = d_{\text{pos}} + d_{\text{time}}$.

$$\mathbf{z} = \mathbf{z}_{\text{pos}} \parallel \mathbf{z}_{\text{time}}$$

where $\parallel$ denotes the vector concatenation operation.

The *Positional Component* $\mathbf{z}_{\text{pos}}$ encodes the relative index P of an event in the sequence using standard Rotary Position Embedding (RoPE). For $\mathbf{z}_{\text{pos}}$, the rotation angles are

$$\theta_j^{\text{pos}}(P) = P \cdot (10000)^{-2j/d_{\text{pos}}}, \quad j \in \{0, \dots, \frac{d_{\text{pos}}}{2} - 1\}.$$

This component disambiguates the order of co-occurring events sharing identical timestamps.

The *Temporal Component* $\mathbf{z}_{\text{time}}$ encodes the timestamp T (in seconds). We utilize a predefined set of M semantically meaningful calendrical periods $\mathcal{S} = \{s_1, s_2, \dots, s_M\}$ (e.g., $s_1=60$ for minute, $s_2=3600$ for hour; see Table 10). For each period $s_m \in \mathcal{S}$, we calculate a rotation angle of timestamp:

$$\theta_m^{\text{time}}(T) = 2\pi \cdot \frac{T \pmod{s_m}}{s_m}, \quad m \in \{1, \dots, M\}.$$

Assuming $d_{\text{time}} = 2M$, the temporal rotation is applied by pairing dimensions. The final embedding $\mathbf{z}'$ is obtained by independently rotating both components:

$$\mathbf{z}' = \text{RoPE}(\mathbf{z}_{\text{pos}}, \Theta^{\text{pos}}(P)) \parallel \text{RoPE}(\mathbf{z}_{\text{time}}, \Theta^{\text{time}}(T))$$

where $\text{RoPE}(\mathbf{y}, \Theta)$ denotes rotating vector $\mathbf{y}$ by the angles in Θ in a pairwise manner, following the RoPE formulation. This allows the attention mechanism to simultaneously address both sequential order and calendrical time (More details in Appendix B.2).

Time-Conditioned Foresee Objective (TCF). We propose a novel learning objective to effectively model the temporal dynamics of EHR data. TCF employs a dual-objective structure that simultaneously learns to: (1) predict *when* the next event will occur ($P(\Delta T_{\text{next}} | E_{\text{past}})$), and (2) foresee *what* events will happen at a specified future times ($P(F_{\text{foresee}} | \Delta T_{\text{foresee}}, E_{\text{past}})$), unlike NTP which only models $P(F_{\text{next}} | E_{\text{past}})$. Figure 2C provides an overview of the following process.

The TCF module is placed after the decoder-only transformer backbone, replacing the original unembedding (output projection) layer. It takes the last hidden state $h_{\text{last}} \in \mathbb{R}^{d_{\text{model}}}$ as input² and outputs both a next time prediction and time conditioned multiple future feature predictions.

To generate a calendrical label of ΔT_{next} , the time delta (in seconds) is transformed into a length N_{scales} integer vector,

²In practice, the full last hidden state $H_{\text{last}} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$ is processed in parallel, as done in NTP—Appendix B.3.

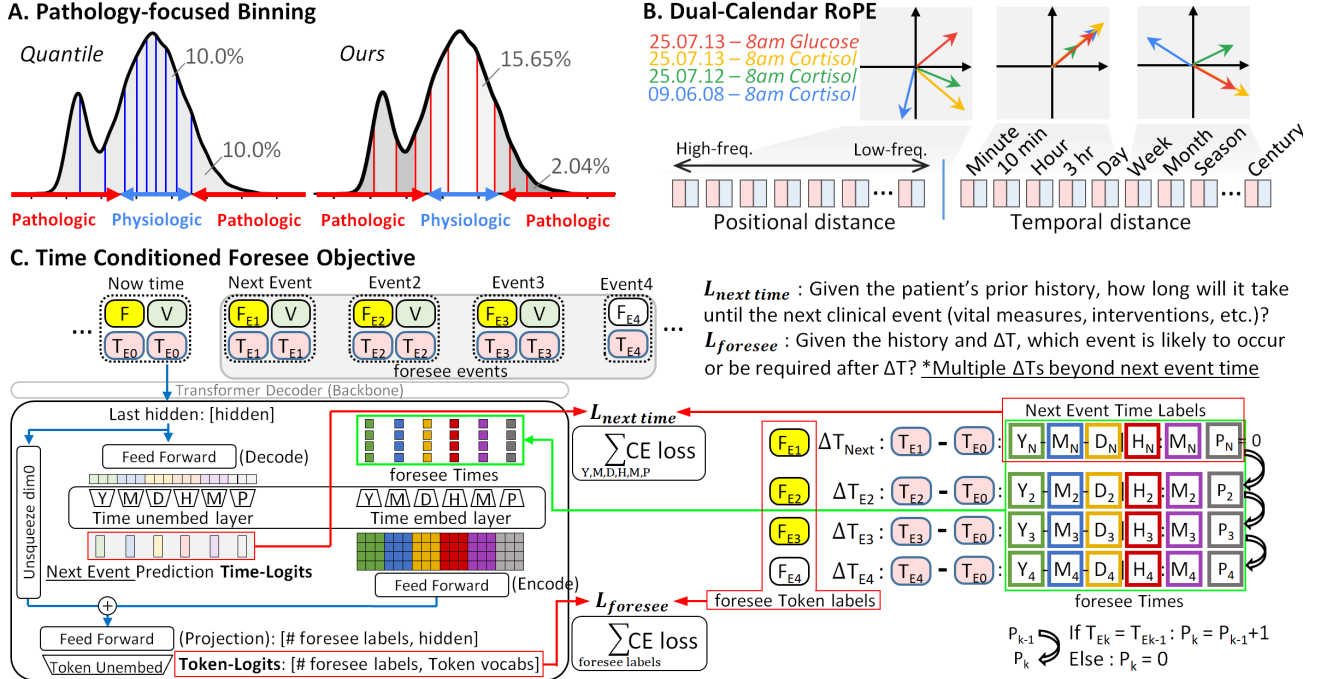


Figure 2. (A) Quantile binning concentrates bins in dense, physiologic ranges. In contrast, our method allocates more bins to sparse pathologic ranges. (B) Concurrent events are distinguished by their positional distance. Events occurring at the same hour on different dates share the same representation for units below a day but have different representations for units of a day or longer. (C) TCF objective comprises $L_{next\ time}$ and $L_{foresee}$. The last hidden state (h_{last}) pass through time head to predict the interval to the next token ($L_{next\ time}$). Then, multiple future intervals are fed back and integrated with the h_{last} to predict corresponding future F s ($L_{foresee}$).

denoted as C_{next} . Each element of this vector corresponds to a predefined calendrical time unit, ranging from 10-year to 1-minute (e.g., vector $[\alpha, \beta, \gamma, \dots]$ represents a time composed of α years, β months, γ days, etc.—see Appendix B.3.1).

To predict C_{next} from h_{last} , the last hidden is projected into N_{scales} vectors of size d_{embed} . Each of these vectors is transformed into time-logit through the unembedding layer.

$$[h_{time}^{(1)} \| h_{time}^{(2)} \| \dots \| h_{time}^{(N_{scales})}] = \text{FFN}_{dec}(h_{last}) \in \mathbb{R}^{(N_{scales} \cdot d_{embed})},$$

$$\text{time-logits}^{(i)} = h_{time}^{(i)} \cdot (W_{embed}^{(i)})^T, \text{ for } i = 1, \dots, N_{scales}.$$

Each $h_{time}^{(i)} \in \mathbb{R}^{d_{embed}}$ corresponds to the latent vector for the i -th time scale. $\mathcal{L}_{next\ time}$ is Cross-Entropy loss between these $\{\text{time-logit}^{(i)}\}_{i=1}^{N_{scales}}$ and the each integer of C_{next} , averaged over N_{scales} .

For foreseeing future events, **Time-Conditioning** is performed. Our goal is to predict the Features for $N_{foresee}$ future events. To this end, the future time deltas $\Delta T_{foresee}$ are transformed into a matrix of integer labels ($C_{foresee} \in \mathbb{Z}^{N_{foresee} \times N_{scales}}$), following the same multi-scale decomposition as ΔT_{next} . These labels are passed through embedding layers to produce a comprehensive time embedding, $e_{time} \in \mathbb{R}^{N_{foresee} \times (N_{scales} \cdot d_{embed})}$. Finally, this time embedding is fused with the original hidden state h_{last} via a residual connection to produce a time conditioned hidden state,

$$h_{conditioned} \in \mathbb{R}^{N_{foresee} \times d_{model}}.$$

$$h_{conditioned} = \text{FFN}_{proj}(\text{LayerNorm}(h_{last} + \text{FFN}_{enc}(e_{time})))$$

This $h_{conditioned}$ is projected to token-logit $\in \mathbb{R}^{N_{foresee} \times \text{vocab-size}}$ that predicts the Features that occur at the corresponding future timestamps.

$\mathcal{L}_{foresee}$ is Cross-Entropy loss between the future Features and the token-logit, averaged over $N_{foresee}$. Through this dual-objective learning ($\mathcal{L} = \mathcal{L}_{next\ time} + \mathcal{L}_{foresee}$), our model acquires the ability to accurately and generatively model a patient's entire medical timeline.

So far, we have considered the position where Feature is predicted given the previous events. Modeling Value given the previous events and Feature is carried out in the same manner. Since F and V belong to the same event and thus share the time label, we always have $\Delta T_{next} = 0$. Moreover, because V is conditioned on the preceding F , we predict V_{now} by modeling

$$P(V_{foresee} | \Delta T_{foresee}, F_{now}, E_{past}), \text{ where } \Delta T_{foresee} = 0$$

Additional details are provided in Appendix B.3.

3.1. Data and Preprocessing

To ensure reproducibility, we use a publicly available dataset and provide open-source code for all stages of the pipeline.

Table 2. Data summary. Parentheses indicate cases where bins are not shared. Refer to Appendix C for more details

Pre-training statistic	MIMIC-III		MIMIC-IV	
	Train	Test	Train	Test
Total Patient #	28.7k	5.1k	159.0k	18.3k
Total Hospitalization #	35.7k	6.3k	455.6k	66.3k
Total Events #	38.6m	6.7m	272.6m	34.3m
Total Tokens #	77.1m	13.5m	534.2m	67.2m
Avg. length	2.7k	2.7k	3.4k	3.7k
Max length	393.3k	62.8k	254.1k	335.7k
Unique Tokens #	155 (1,208)		361 (2301)	
Token # bin	10		10	
Token # ethnicity	10		7 (grouped)	
Token # chart events	17		148	
Token # laboratory tests	100		120	
Token # other features	0		9	

We employ the MIMIC-III v1.4 and MIMIC-IV v3.1 Clinical Database (Johnson et al., 2016a; 2023; 2024). MIMIC-III focuses on ICU stays, whereas MIMIC-IV additionally includes emergency department and general ward. Together, these datasets provide a comprehensive testbed covering diverse clinical settings and varying data density. We adopt the widely used preprocessing pipeline (Harutyunyan et al., 2019). To demonstrate that our Dual-calendar RoPE generalizes to temporal shifts, we constructed a test set on MIMIC-IV consisting of the latest 10% of patients by birth date. This split induces an average temporal shift of 41 years. A dataset summary is provided in Table 2. Further details are provided in Appendix C. Tasks necessitating clinical judgment, e.g., defining exclusion criteria and outlier removal, were independently reviewed by three physicians.

3.2. Backbone, Baseline models, and Pre-training

We used the same Transformer decoder as the backbone for all experiments. In case of MLM model, encoder was used for pre-training and causal mask was applied for downstream tasks. For Dual-calendar RoPE, the first 24 dimen-

sions of the 64-dim K and Q vectors encode positional information, and the remaining 40 dimensions encode calendric time. Baseline models were reproduced under identical conditions, including backbone and training data. We followed the original papers’ implementations but made necessary modifications where direct application was infeasible (e.g., adapting the MOTOR model to numeric value events). The pre-training input token length was fixed at 2048. Sequences exceeding this length (Figure 6) were segmented with a 512-token overlap. A detailed description of our model, baselines, and pre-training can be found in Appendix D.

4. Result

4.1. Downstream task and Fine-tuning

Wide range of clinical downstream tasks were applied. These tasks are defined by clinical labels excluded from training and serve as proxies for the quality of patient representations. In addition to the four MIMIC-III benchmark (Harutyunyan et al., 2019) tasks—In-hospital Mortality (IHM), Decompensation-death (Dec-death), Length of Stay (LOS-ICU), and Phenotyping (Phe)—we included three additional tasks: Decompensation-arrest (Dec-arrest), Oliguria/Anuria (HUO), and Vasopressor (Vaso) use.

For MIMIC-IV, we further incorporated four tasks that reflect general ward (GW) and post-discharge outcomes: Decompensation-ICU (Dec-ICU), which predicts ICU transfer 24 hours in advance while patients are in the GW; Length of Stay including GW (LOS-Adm.); and 30-day readmission and mortality prediction at discharge (30d-Readm. and 30d-death). Label counts for all tasks are provided in Appendix Table 13, with detailed descriptions in Appendix E.1.

For MIMIC-III, to evaluate generalization to data with different distributions (e.g., missing lab information), we ex-

Table 3. Performance on downstream tasks using EHR datasets with 117, 17, and 6 features. The ‘Total score’ denotes a weighted cross-entropy objective computed over the test sets of all downstream tasks, weight adjusted for class imbalance (lower is better). For the 117-feature setting, we report results on seven tasks including IHM and Vaso. Binary classification tasks are evaluated using AUROC and AUPRC, while multiclass tasks are evaluated using Macro F1 and Cohen’s Kappa. For tasks with multiple subtasks, both macro and micro AUROC are reported. Our model, trained both with and without value sharing, consistently outperforms all baselines. Full results are provided in Appendix Tables 14–16.

Tasks		Total score ($\downarrow$)			IHM		Phe		Dec-death		Dec-arrest		LOS-ICU		HUO		Vaso	
Metric		117	17	6	ROC	PRC	macro	micro	ROC	PRC	ROC	PRC	Ma-f1	Kappa	macro	micro	ROC	PRC
No Value share	HEART	5.304	5.434	5.835	0.838	0.442	0.717	0.718	0.869	0.205	0.862	0.199	0.150	0.142	0.703	0.701	0.865	0.363
	MOTOR	4.889	5.159	5.645	0.880	0.552	0.770	0.773	0.908	0.274	0.896	0.268	0.177	0.166	0.753	0.748	0.891	0.438
	EHRSHOT	5.497	5.565	6.026	0.831	0.459	0.638	0.632	0.867	0.188	0.854	0.181	0.147	0.155	0.672	0.668	0.54	0.359
	TRADE	5.260	5.454	6.048	0.828	0.441	0.738	0.738	0.867	0.170	0.857	0.165	0.158	0.151	0.732	0.731	0.869	0.393
	EHRmamba	5.137	5.439	5.926	0.868	0.557	0.690	0.687	0.901	0.277	0.886	0.260	0.150	0.159	0.751	0.753	0.873	0.399
	Ours (No share)	4.686	4.907	5.367	0.889	0.607	0.809	0.816	0.928	0.400	0.917	0.388	0.181	0.185	0.776	0.781	0.912	0.498
Value share	FM4EHR	5.529	5.691	6.200	0.830	0.435	0.609	0.599	0.886	0.273	0.874	0.261	0.131	0.153	0.658	0.667	0.878	0.384
	ETHOS	4.971	5.248	5.572	0.859	0.530	0.739	0.746	0.900	0.311	0.890	0.304	0.170	0.165	0.721	0.731	0.890	0.437
	STraTS	5.786	5.812	6.071	0.759	0.311	0.656	0.661	0.840	0.141	0.804	0.103	0.123	0.121	0.590	0.598	0.864	0.331
	Ours (Share)	4.879	5.043	5.561	0.876	0.559	0.781	0.784	0.910	0.319	0.902	0.310	0.173	0.170	0.749	0.755	0.906	0.470

Table 4. Performance comparison against baselines on the MIMIC-IV dataset. We compare ‘No Share’ models (MOTOR vs. Ours) and ‘Share’ models (ETHOS vs. Ours). The best results are highlighted in bold.

Task	Metric	No Share		Share	
		MOTOR	Ours	ETHOS	Ours
IHM	ROC	0.908	0.916	0.876	0.894
	PRC	0.385	0.419	0.284	0.336
Dec-death	ROC	0.944	0.953	0.922	0.952
	PRC	0.188	0.302	0.162	0.278
Dec-arrest	ROC	0.937	0.947	0.918	0.947
	PRC	0.183	0.281	0.154	0.259
Dec-ICU	ROC	0.845	0.863	0.786	0.862
	PRC	0.160	0.213	0.116	0.233
30d-Readm.	ROC	0.659	0.668	0.611	0.667
	PRC	0.365	0.381	0.311	0.389
30d-death	ROC	0.879	0.892	0.818	0.881
	PRC	0.309	0.330	0.145	0.296
LOS-Adm.	Ma-F1	0.181	0.188	0.158	0.184
	Kappa	0.184	0.193	0.154	0.190
LOS-ICU	Ma-F1	0.190	0.200	0.161	0.192
	Kappa	0.173	0.183	0.126	0.166
Phe.	macro	0.816	0.840	0.719	0.816
	micro	0.829	0.852	0.724	0.825
Vaso	ROC	0.963	0.971	0.952	0.959
	PRC	0.691	0.732	0.602	0.657
HUO	macro	0.760	0.822	0.700	0.754
	micro	0.761	0.823	0.702	0.754

Table 5. Cross-hospital generalization on the external eICU dataset. Settings as Table 4. To assess robustness against temporal distribution shifts, models are fine-tuned on patients from 2014 and evaluated on those from 2015. The best results are in bold.

Task	Metric	No Share		Share	
		MOTOR	Ours	ETHOS	Ours
IHM	ROC	0.757	0.786	0.735	0.771
	PRC	0.324	0.355	0.298	0.346
ICUM	ROC	0.779	0.811	0.765	0.795
	PRC	0.256	0.286	0.234	0.284

perimented with three input configurations: all 117 features, 17 vital signs (without lab data), and only 6 vital signs (SBP, DBP, body temperature, heart rate, respiratory rate, SpO2). For MIMIC-IV, we used a richer set of features drawn from `chartevents`, `labevents`, `OMR`, and `outputevents`. Based on event frequency, we initially selected 250 features and merged semantically equivalent features (e.g., weight in lb vs. kg), yielding 195 features. Please refer to Appendix E.2 for more details.

Table 3 reports downstream task results on MIMIC-III. For a fair comparison, we evaluate our model both with and without bin sharing. In both settings, our model outperforms all baselines across the three input configurations. Except for the IHM task, these improvements are statistically significant compared to the second-best model ($p < 0.05$).

Table 6. Comparison with ML models on the IHM task. MIMIC-III uses 117 features and MIMIC-IV uses 195 features, where feature values are computed by averaging measurements up to 48 hours after admission. Values in parentheses denote the 95% confidence interval, and the best results are highlighted in red.

Model	AUROC (95% CI)	AUPRC (95% CI)
<i>Dataset: MIMIC-III (117 features)</i>		
Logistic Regression	0.780 (0.752–0.803)	0.392 (0.342–0.447)
Decision Tree	0.634 (0.609–0.658)	0.190 (0.165–0.215)
Random Forest	0.854 (0.834–0.872)	0.501 (0.448–0.549)
XGBoost (2016)	0.860 (0.841–0.879)	0.509 (0.456–0.566)
LightGBM (2017)	0.881 (0.866–0.896)	0.556 (0.510–0.603)
CatBoost (2018)	0.883 (0.869–0.900)	0.565 (0.522–0.614)
MOTOR	0.880 (0.855–0.890)	0.552 (0.500–0.591)
OURS	0.889 (0.874–0.905)	0.607 (0.566–0.649)
<i>Dataset: MIMIC-IV (195 features)</i>		
Logistic Regression	0.848 (0.836–0.859)	0.256 (0.234–0.282)
Decision Tree	0.607 (0.596–0.619)	0.076 (0.068–0.086)
Random Forest	0.878 (0.868–0.889)	0.321 (0.295–0.352)
XGBoost	0.892 (0.884–0.901)	0.344 (0.319–0.371)
LightGBM	0.900 (0.893–0.910)	0.343 (0.314–0.375)
CatBoost	0.906 (0.898–0.914)	0.359 (0.334–0.387)
MOTOR	0.908 (0.899–0.914)	0.385 (0.365–0.407)
OURS	0.916 (0.908–0.923)	0.419 (0.391–0.446)

Notably, on the decompensation task—predicting death or arrest up to 24 hours in advance—our model achieves an AUPRC 45% higher than the second-best model. Given the severe class imbalance (positive:negative ratio of 1:40), this gain is particularly meaningful for real-world clinical applications where high precision is critical.

Table 4 presents downstream task results on MIMIC-IV, comparing our method with the second-best models from MIMIC-III. Despite the richer dataset including general ward and emergency department data, our approach consistently outperforms all baselines across tasks, with an average improvement of 18.0% across metrics. Notably, it achieves substantially higher AUPRC scores (40.7% on average) under severe class imbalance, underscoring its practicality in real-world clinical settings.

External validation on independent hospitals is essential to demonstrate general-purpose capability. As shown in Table 5, we evaluate on the eICU dataset (Pollard et al., 2019), which has a distinct distribution from MIMIC, comparing against the strongest baseline in each group (MOTOR for No Share, ETHOS for Share) on In-Hospital Mortality (IHM) and ICU Mortality (ICUM) prediction. Fine-tuning on 2014 patients and evaluating on 2015 to test temporal robustness, our model consistently outperforms the baselines across both tasks and settings.

We benchmarked our model against ML baselines traditionally superior to deep learning for tabular data. As shown in Table 6, our proposed method consistently outperformed all others, sustaining a dominant AUPRC across all tasks.

Table 7. Ablation study on Pathology-Focused Binning (Binning), Dual-Calendar Rotary Positional Embedding (Embedding), and Time-Conditioned Foreseeing (Objective).

Binning	Embedding	Objective	Total score	IHM-ROC	IHM-PRC
V	V	V	4.686	0.889	0.607
Quantile	V	V	4.713	0.880	0.583
Quantile	RoPE	V	4.810	0.877	0.519
Quantile	RoPE	NTP	5.241	0.837	0.448

Table 8. TCF Objective Ablation. TCF objective consists of (a) $L_{\text{next_time}}$ and L_{foresee} , where L_{foresee} predicts (b) **multiple future tokens** (c) **conditioned on their occurrence times**. This analysis is performed with half training steps.

Ablation conditions	Total score ($\downarrow$)
Time-Conditioned Foresee (a, b, c)	4.762
Remove $L_{\text{next_time}}$ (b, c)	4.816
Single token (next token) for L_{foresee} (a, c)	4.844
Multiple tokens w/o time conditioning (a, b)	4.839
Simple multiple-token prediction (b)	4.886
NTP (None)	5.054

Ablation study on the three components (Table 7) shows that each contributes substantially to the performance improvement. More detailed results on downstream tasks can be found in Appendix F.1. To isolate the source of TCF’s effectiveness, we conduct a fine-grained ablation on its components (Table 8). Removing the interval prediction, replacing multi-token foreseeing with a single next token, or dropping time conditioning each degrades the Total score. This confirms that the gains stem specifically from time-conditioning future tokens rather than merely predicting more of them.

We further provide extensive analyses in Appendix F.2, including a baseline capacity study (showing that increasing baseline depth yields negligible gains and does not close the gap to our model) and component-wise hyperparameter analyses for Pathology-Focused Binning (weighting λ and bin count) and Dual-Calendar RoPE (number of calendrical components). These results confirm that our improvements stem from the proposed architectural choices rather than model capacity, and that each component is robust around its default setting.

4.2. Temporal generative modeling

Our model is the first to generate fine-grained timestamps and incorporate long-range supervision. We compared our model against ETHOS—one of the few baselines capable of generating temporal information, despite its limitations in precision. Since ETHOS outputs quantized interval tokens, its timestamps were sampled and rounded to the nearest five minutes to align with the MIMIC’s resolution. For both models, bin indices were decoded by sampling from the training data’s empirical distributions.

# [Ours (bin share)]	# [ETHOS]
Birth : 2055. 03. 02	Birth : 2055. 03. 02
Sex: Female	Sex: Female
Ethnicity: Asian	Ethnicity: Asian
Age: 81	Age: 81
2136-10-02 13:25:04	2136-10-02 13:25:04
- ICU transfer	- ICU transfer
2136-10-02 13:49:59	2136-10-02 13:49:59
- RR : 12	- RR : 12
2136-10-02 14:00:00	2136-10-02 14:00:00
- RR : 18	- RR : 18
- SBP : 118	- SBP : 118
- O2 saturation : 99	- O2 saturation : 99
- MBP : 77	- MBP : 77
- DBP : 51	- DBP : 51
- GCS-V : 1	- GCS-V : 1
- GCS : 3	- GCS : 3
- GCS-M : 1	- GCS-M : 5
- Temperature : 38.3	- GCS-E : 2
- GCS-E : 1	- Temperature : 38.2
2136-10-02 14:04:00	- HR : 74
- Potassium (ER) : 3.5	2136-10-02 14:50:00
- PO2 : 492.0	- RR : 15
- PEEP : 5.0	- SBP : 96
- CO2 : 26.0	- O2 saturation : 100
- pH : 7.44	- MBP : 89
- Base excess : 1.0	- DBP : 50
- Hemoglobin (ER) : 10.1	- HR : 78
- PCO2 : 37.0	2136-10-02 15:35:00
2136-10-02 14:10:00	- RR : 28
- SBP : 104	- O2 saturation : 96
- RR : 15	2136-10-02 16:35:00
- O2 saturation : 99	- GCS : 9
- MBP : 69	- GCS-E : 1
- DBP : 45	- DBP : 80
2136-10-02 15:00:00	- GCS-M : 3
- SBP : 125	- O2 saturation : 96
- GCS : 3	- MBP : 86
- O2 saturation : 100	- HR : 76
- HR : 80 (truncated rest)	- GCS-V : 1 (truncated rest)

Figure 3. Given the initial record (orange), the subsequent medical history is generated (blue). PEEP: Positive end-expiratory pressure, ER: emergency lab.

Figure 3 presents generated medical history sequences from our model and ETHOS, given the same initial EHR records. From a content perspective, the Glasgow Coma Scale (GCS) should equal the sum of GCS-E/V/M. At 10-02 14:00, our model generated E:1, V:1, M:1; GCS:3, correctly capturing this relationship, whereas ETHOS produced E:2, V:1, M:5; GCS:3, which is inconsistent. Moreover, our model reflected early emergency labs and a variety of tests, followed by routine vital sign checks, while ETHOS generated no labs. From a temporal perspective, our model first performed several tests at short intervals after admission, then naturally returned to an hourly routine. In contrast, ETHOS produced events at irregular intervals and often failed to follow the typical hourly schedule.

Table 9 quantifies generative capability along the two components of the next event: (a) the interval to the next event and (b) the token of the next event. For timestamps, our model reduces RMSE/MAE by up to 99% over ETHOS, with the largest gains on clinically crucial hour-scale intervals where ETHOS’s coarse interval tokens accumulate error. For token content (z-normalized), our model is consistently more accurate across the value range, improving on both

Table 9. Quantitative evaluation of generative capability. Generating the next event requires predicting both (a) the interval to the next event and (b) its content. We compare against ETHOS; the improvement columns report the relative error reduction (%). All token values in (b) are z -normalized.

(a) Timestamp Generation						
Interval (Minutes)	ETHOS		OURS		Improv. (%)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
0–1	13426	163	140	1.6	99	99
1–10	13913	244	1077	11	92	95
10–100	50409	2404	645	31	99	99
100–1k	61858	7939	1796	331	97	96
(b) Token Generation						
Token (Normalized)	ETHOS		OURS		Improv. (%)	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
$ \sigma < 1\text{ sd}$	0.520	0.307	0.438	0.265	16	14
$ \sigma > 2\text{ sd}$	2.127	1.279	1.972	1.170	7	9
Overall	0.680	0.368	0.623	0.337	8	9

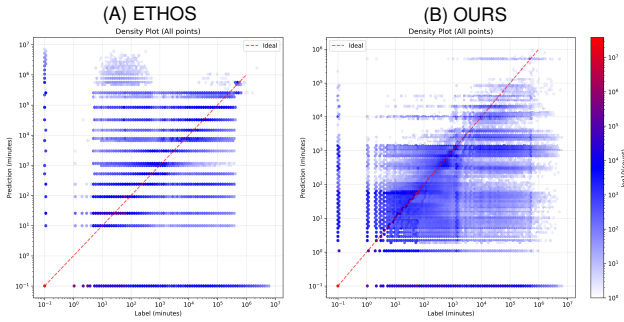


Figure 4. Visualizing EHR generation time interval performance. This figure illustrates the relationship between the actual next-token interval (x -axis) and the model-generated interval (y -axis) for (A) ETHOS and (B) our model.

near-mean ($|\sigma| < 1\text{sd}$) and far-from-mean ($|\sigma| > 2\text{sd}$) measurements. This indicates that Pathology-Focused Binning sharpens resolution in the clinically critical pathological range without sacrificing fidelity in the dense central region. Figure 4 further visualizes this gap in the generated time intervals, confirming that our approach captures both *when* and *what* more faithfully than the ETHOS temporal-generative baseline.

We further evaluated 100 generated samples with three evaluator groups: five physicians, five non-medical participants, and four commercial LLMs. After reviewing up to 10 ground-truth EHR samples, each group assessed subsequent EHR records generated from the same initial records. Figure 5 shows that our model consistently outperformed ETHOS. The LLM input prompts and the generated samples are presented in Appendix F.3.

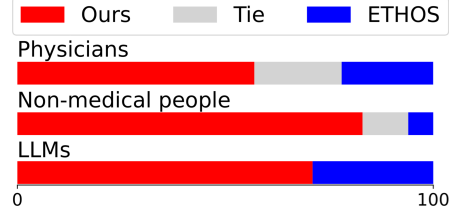


Figure 5. The results represent quality assessments across three groups: physicians ($n=5$), non-medical participants ($n=5$), and LLMs (GPT-4.5, Gemini 2.5 Flash/Pro, and Claude 4 Sonnet via API). For each group, 100 pairwise comparison responses were collected and analyzed.

5. Conclusion

We present an approach for modeling the unique characteristics of EHR data, addressing irregular event intervals and complex numerical values that distinguish clinical records from natural language. Our contributions include Pathology-Focused Binning, which allocates finer resolution to clinically significant value ranges; Dual-Calendar RoPE, which jointly encodes relative and absolute calendrical time; and the Time-Conditioned Foreseeing (TCF) objective, which predicts future timestamps and events in alignment with how clinicians plan care. Our model outperforms existing foundation models on diverse downstream tasks with up to 54% AUPRC improvement, while generating realistic, temporally consistent EHRs for long-range forecasting. We further confirm cross-hospital generalization on the external eICU dataset, where our model outperforms the strongest baselines on In-Hospital and ICU Mortality prediction and remains robust under year-to-year temporal distribution shifts.

Limitations: Currently, limited metrics exist for evaluating the temporal generative performance of EHR models. Assessing the appropriateness of event timing is crucial yet difficult to capture with conventional metrics developed for LLM evaluation. Developing quantitative measures tailored to EHR generation will be important for reliably comparing and advancing EHR foundation models. Our experiments are also confined to ICU-centric datasets, and broader validation across other clinical settings would help characterize the scope of these findings.

Future Work: A promising direction is integrating our approach with existing LLMs and foundation models, mitigating the “token explosion” of structured-to-text conversion via lightweight projection or alignment layers. Our Dual-Calendar RoPE is well-suited to such hybrid settings, as it decouples an event’s recording time from its token position, while the TCF objective remains natively applicable. We see extending these ideas to multimodal records that combine structured events with clinical notes as a natural next step.

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Impact Statement

This research introduces a specialized foundation model architecture for Electronic Health Records (EHRs) that bridges the gap between general-purpose machine learning and the nuanced demands of clinical decision-making. The potential broader impacts of this work on society and the medical field are as follows:

- **Enhancement of Clinical Safety:** By introducing Pathology-Focused Binning, our model provides higher resolution for sparse but clinically critical pathological ranges. This directly improves the detection and prediction of high-risk medical events that are often overlooked by standard quantile-based models.
- **Long-range Clinical Planning:** Unlike traditional short-sighted objectives, our Time-Conditioned Foreseeing (TCF) objective aligns with the way clinicians actually plan treatments by forecasting future events across multiple temporal horizons. This advancement supports more proactive and comprehensive patient management in intensive care units (ICUs) and general wards.
- **High-Fidelity Synthetic Trajectories:** The model’s ability to generate realistic, temporally consistent patient trajectories offers a significant breakthrough for medical education and research. This enables the development of robust AI tools even in data-scarce environments while maintaining high standards of clinical plausibility as validated by medical professionals.
- **Ethical Considerations and Fairness:** We emphasize that this model is intended to augment, not replace, clinical judgment. Our study involved rigorous review by

multiple physicians to ensure clinical integrity. Furthermore, we contribute to the open-science community by open-sourcing our code to facilitate transparency and independent validation of medical AI.

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A. Additional Related Works

The pursuit of powerful and versatile foundation models for Electronic Health Records (EHRs) has led to a rapid evolution of modeling techniques. Current models draw heavily from advancements in Natural Language Processing (NLP), treating EHR data as sequences of discrete events. However, the unique characteristics of EHRs—specifically their sparse, irregularly sampled nature, and the continuous numerical values—have necessitated the development of more specialized architectures and learning objectives.

Early Approaches: Modeling Categorical Event Sequences

The initial wave of EHR foundation models adapted the successful Transformer architecture from NLP to the clinical domain. These models primarily focused on learning representations from sequences of medical codes, such as diagnoses, procedures, and medications, while largely omitting numerical and temporal data.

BEHRT (Li et al., 2020) (Transformer for Electronic Health Records) introduced the use of the Bidirectional Encoder Representations from Transformers (BERT (Devlin et al., 2019)) architecture for EHR data. It treats a patient’s EHR as a sequence of “sentences,” where each sentence is a collection of medical codes from a single visit. *BEHRT* is pre-trained on a large dataset of patient records using a Masked Language Model (MLM) objective, where the model learns to predict masked medical codes based on their context. An additional task, Next Visit Prediction (NVP), was also used to predict the codes for a subsequent visit. While effective for tasks like disease prediction, *BEHRT*’s exclusion of numerical values and timestamps limits its clinical utility, as it cannot capture disease severity or the precise timing of events.

Med-BERT (Rasmy et al., 2021) followed a similar approach to *BEHRT*, applying the BERT architecture to structured EHR data. It also represents patient histories as sequences of medical codes and uses an MLM pre-training objective to learn contextualized embeddings. *Med-BERT* demonstrated strong performance on various downstream tasks, including disease prediction and patient mortality prediction. However, like *BEHRT*, it does not explicitly model the temporal intervals between visits or the continuous values of lab tests, which are crucial for a comprehensive understanding of a patient’s health trajectory.

EHR-BERT (Niu et al., 2024) is another BERT-based model that focuses on detecting anomalies in EHR data. It learns the typical patterns of medical events and flags deviations from these patterns as potential anomalies. While its primary application is in data quality and fraud detection, it shares the same fundamental limitations as other early BERT-based models in its handling of EHR data, as it does not incorporate numerical or temporal information into its core architecture.

Foresight (Kraljevic et al., 2024) relies on a GPT-2 backbone with standard learned positional embeddings. While the model incorporates temporal context by inserting age tokens and distinguishing daily intervals with separator tokens (<SEP>), it does not assign precise timestamps to individual events. Instead, *Foresight* aggregates events into daily buckets, effectively treating all events within the same day as sharing identical temporal information. Because this approach flattens the intra-day temporal dimension and obscures specific event timings, we classify *Foresight* as a method that does not utilize event timestamps.

ClinicalMamba (Yang et al., 2024) established the viability of large-scale pre-training for EHR data and demonstrated the power of the Transformer architecture in capturing the complex relationships between medical events. However, their reliance on a purely categorical representation of patient histories highlighted the need for more sophisticated methods that could incorporate the rich numerical and temporal information present in EHRs.

Incorporating Numeric Values

Recognizing the limitations of purely categorical models, subsequent research focused on integrating continuous numerical values, such as lab results and vital signs, into the modeling process. A common approach has been to discretize these values into a set of predefined bins, allowing them to be treated as discrete tokens within the existing language modeling framework.

HEART (Huang et al., 2024) employs this discretization strategy. They convert numeric values into a fixed number of uniform bins (e.g., 10 bins) and create a unique token for each “feature-value” pair. This allows them to capture the magnitude of numerical measurements to some extent. However, this approach has two major drawbacks. First, it leads to a massive increase in the vocabulary size, as each feature needs its own set of value-specific tokens. This exacerbates data sparsity issues and increases the model’s memory footprint. Second, uniform binning can be suboptimal, as it focuses on the

non-significant range of clinical values and may not provide sufficient resolution for clinically significant changes.

FM4EHR (Burkhart et al., 2025) (Foundation Models for Electronic Health Records) proposed a more efficient method for handling numeric values. Instead of creating unique tokens for each feature-value pair, *FM4EHR* separates the tokenization of features and values. This allows different features to share the same set of value tokens, significantly reducing the vocabulary size and mitigating the data sparsity problem. This “bin sharing” (bin token sharing) approach is a key innovation that allows for more scalable and efficient modeling of numerical data. However, *FM4EHR* still does not explicitly model the temporal aspect of EHR data, relying on the implicit ordering of events in the sequence.

Addressing Temporal Information

The timing of medical events is often as important as the events themselves. Another line of research has focused on developing models that can explicitly encode temporal information of EHR data.

MOTOR (Steinberg et al., 2024) (A Time-to-Event Foundation Model for Structured Medical Records) is a model specifically designed for survival analysis and time-to-event prediction. It processes sequences of medical codes and learns to predict the time to a future event of interest. (It does not take numerical values) *MOTOR* represents time by discretizing the time horizon into a set of predefined intervals and models the hazard function within each interval. This allows it to capture the temporal dependencies between events and make time-aware predictions. However, *MOTOR*’s primary limitation is that it does not incorporate numerical values, which are often strong predictors of patient outcomes. Additionally, its reliance on predefined time intervals and the assumption of a constant hazard function within each interval can limit its temporal precision.

STraTS (Tipirneni & Reddy, 2022) (Self-Supervised Transformer for Sparse and Irregularly Sampled Multivariate Clinical Time-Series) takes a different approach to modeling time and values. It tokenizes only the categorical features and embeds time intervals and numerical values as continuous variables, where temporal information is transformed into embeddings via a feed-forward network, using absolute positional embeddings along the time dimension. *STraTS* is trained using a Mean Squared Error (MSE) loss to predict the values of different features at future time points. This allows it to handle irregularly sampled data and make fine-grained predictions. However, by only predicting the next 2-hour value, *STraTS* loses important contextual information and cannot be used for generative modeling of entire patient trajectories.

EHRSHOT (Wornow et al., 2023) introduces a longitudinal benchmark and releases the weights for CLMBR-T-base, a causal transformer trained on the structured EHRs of 2.57 million patients. CLMBR-T-base operates with minute-level resolution to capture precise temporal sequencing. To model time, it applies both absolute positional embedding and Rotary Positional Embeddings (RoPE), which are added to the code embeddings to effectively capture relative temporal distances between clinical events. Regarding numerical data, *EHRSHOT* adopts a discretization strategy where lab values are converted into categorical bins based on decile statistics computed over the dataset. While this enables the unified processing of values and codes via a standard autoregressive objective, the fixed vocabulary size necessitates discarding less frequent codes and limiting the granularity of numerical representations.

TRADE (Zhu et al., 2024) (Predicting Risk of Alzheimer’s Diseases and Related Dementias with AI Foundation Model on Electronic Health Records) also incorporates numerical values, and it uses a non-uniform binning. It discretizes values into nine bins based on their standard deviation from the mean. This approach is more clinically plausible than uniform binning, as it can better capture extreme values that are often indicative of disease. However, *TRADE* does not employ bin token sharing, which means it still faces the challenge of a large and sparse vocabulary.

EHRMamba (Fallahpour et al., 2025) is a recent model that leverages the Mamba architecture, a type of State Space Model (SSM), to efficiently process long EHR sequences. It tokenizes categorical features and uses uniform binning for numerical values. To capture temporal dependencies, it utilizes the Time2Vec module (Kazemi et al., 2019) to encode patient age. Notably, this formulation functions as an absolute positional embedding, analogous to the sinusoidal embeddings in the original Transformer. While this yields a periodic representation of time, its reliance on absolute positioning renders it vulnerable in terms of generalization compared to relative encoding schemes. The use of the Mamba architecture allows *EhrMamba* to scale to much longer patient histories than Transformer-based models, which have a quadratic complexity with respect to sequence length.

ETHOS (Renc et al., 2024) (Zero-shot health trajectory prediction using transformer) introduces a novel method for explicitly modeling the time intervals between events. It discretizes the time gaps into 13 logarithmic bins, ranging from minutes to months, and inserts a special “time token” between each event token in the input sequence. This allows the model to

explicitly reason about the temporal relationships between events. *ETHOS* also incorporates numerical values through binning and shares bins across features. While this explicit time tokening is a significant step forward, it can increase the sequence length and computational cost. Moreover, the discretization of time still imposes a limit on the model’s temporal precision.

B. Additional Method

B.1. Pathology-focused Binning

This numerical value tokenization method is designed to create a granular representation for clinically important pathologic values. This non-parametric approach assigns greater resolution to sparse, low density value ranges without making distributional assumptions. The process consists of two main stages: (1) value weight assignment via Kernel Density Estimation, and (2) weighted percentile binning using these assigned weights.

B.1.1. VALUE WEIGHT ASSIGNMENT VIA KERNEL DENSITY ESTIMATION

The core principle is to assign low weights to values in high-density (physiologic) regions and high weights to values in low-density (pathologic) regions. This is achieved by estimating the data density for each medical feature and assigning a weight inversely proportional to this density.

For a feature with a set of values V and standard deviation σ , we define a set of discrete representative points $X = \{x_1, x_2, \dots, x_M\}$. These points span the feature’s range $[\min(V), \max(V)]$ and are spaced at uniform intervals of 0.05σ .

At each discrete point $x_k \in X$, we estimate the data density $\rho(x_k)$ by applying a Gaussian convolution kernel. This is a form of Kernel Density Estimation (KDE), where the density at x_k is the sum of influences from all unique data values $v_j \in V$. The density is:

$$\rho(x_k) = \sum_{j=1}^{|V|} K_h(x_k - v_j)$$

Here, $K_h(u)$ is an unnormalized Gaussian kernel defined as:

$$K_h(u) = \exp\left(-\frac{u^2}{2h^2}\right)$$

The bandwidth h , which controls the smoothness of the density estimate, is set to $h = 0.1\sigma$ to capture local variations.

From this density, we calculate a raw weight $w_{\text{raw}}(x_k) = 1/(\rho(x_k) + \epsilon)$ for each discrete point. These weights are then normalized and clipped to produce the final weight:

$$w_{\text{final}}(x_k) = \min\left(\frac{w_{\text{raw}}(x_k)}{\min_{k'} w_{\text{raw}}(x_{k'})}, w_{\text{max}}\right)$$

where w_{max} is a predefined ceiling (e.g., 10). Finally, each original unique value v_j is assigned the weight of its nearest discrete point, $w(v_j) = w_{\text{final}}(x_{k^*})$, where $k^* = \arg \min_k |v_j - x_k|$.

B.1.2. WEIGHT CALCULATION AND NORMALIZATION

With weights assigned, we proceed to the binning stage. The goal is to partition the feature’s values into B bins such that regions with higher weights are given more bins.

We start by calculating a **weighted count** c'_j for each unique value v_j :

$$c'_j = c_j \cdot w(v_j)$$

This new count reflects the value’s clinical importance as determined by its rarity. Next, we compute the total weighted count for the feature, $C'_{\text{total}} = \sum_{j=1}^{|V|} c'_j$.

The bin thresholds are then determined from the cumulative distribution of these weighted counts. For a set of sorted unique values $v_1 < v_2 < \dots < v_{|V|}$, the cumulative weighted count up to value v_k is $S_k = \sum_{j=1}^k c'_j$. The threshold for the p -th

bin (where $p \in \{1, 2, \dots, B - 1\}$) is set to the first value v_k whose cumulative weighted share meets or exceeds the p/B percentile:

$$T_p = \min\{v_k \mid \frac{S_k}{C'_{\text{total}}} \geq \frac{p}{B}\}$$

This procedure ensures that value ranges containing high-weight (pathologic) data contribute more significantly to the cumulative sum. As a result, a smaller span of these values is needed to cross a percentile boundary, leading to a denser allocation of bin thresholds in these clinically important regions.

Algorithm 1 Density-Based Value Weight Assignment

Require:

item_counters: A map from item ID $\rightarrow$ {value: count}.
 w_{max} : Maximum weight threshold (e.g., 10).

Ensure:

value_weights: A map from item ID $\rightarrow$ {value: weight}.

```

0: Initialize value_weights  $\leftarrow \emptyset$ 
0: for all item_id, counter in item_counters do
0:    $V \leftarrow$  sorted unique values from counter
0:    $C \leftarrow$  corresponding counts for each value in  $V$ 
0:    $AllValues \leftarrow$  list of all values repeated by their counts
0:    $\sigma \leftarrow \text{StandardDeviation}(AllValues)$ 
0:   if  $\sigma = 0$  then
0:     item_weights  $\leftarrow \{v : 1.0 \text{ for } v \in V\}$ 
0:     value_weights[item_id]  $\leftarrow$  item_weights
0:   continue
0:    $h \leftarrow 0.1 \times \sigma$  {Set bandwidth for the Gaussian kernel}
0:    $interval \leftarrow 0.05 \times \sigma$ 
0:    $SplitPoints \leftarrow$  Generate points from  $\min(V)$  to  $\max(V)$  with  $interval$ 
  {Step 1: Kernel Density Estimation}
0:   densities  $\leftarrow \emptyset$ 
0:   for all split point  $x$  in  $SplitPoints$  do
0:      $\rho(x) \leftarrow \frac{1}{|V|} \sum_{j=1}^{|V|} C_j \cdot \exp\left(-\frac{(x-V_j)^2}{2h^2}\right)$ 
0:     densities.append( $\rho(x)$ )
  {Step 2: Calculate, Normalize, and Clip Weights}
0:    $w_{raw} \leftarrow 1.0 / (\text{densities} + 10^{-10})$ 
0:    $w_{norm} \leftarrow w_{raw} / \min(w_{raw})$ 
0:    $w_{final} \leftarrow \text{clip}(w_{norm}, 1.0, w_{max})$ 
  {Step 3: Assign weights to original values}
0:   item_weights  $\leftarrow \emptyset$ 
0:   for all value  $v$  in  $V$  do
0:      $closest\_idx \leftarrow \arg \min_k |SplitPoints_k - v|$ 
0:     item_weights[v]  $\leftarrow w_{final}[closest\_idx]$ 
0:   value_weights[item_id]  $\leftarrow$  item_weights
0: return value_weights=0
    
```

Algorithm 2 Weighted Percentile Binning

Require:

item_counters: A map from item ID $\rightarrow$ {value: count}.
 value_weights: The output from Algorithm 1.
 B: The desired number of bins (e.g., 100).

Ensure:

bin_thresholds: A map from item ID $\rightarrow$ a list of $B - 1$ thresholds.

```

0: Initialize bin_thresholds  $\leftarrow \emptyset$ 
0: for all item_id, counter in item_counters do
0:    $V \leftarrow$  sorted unique values from counter
0:    $N \leftarrow |V|$ 
0:   {Apply weights to counts for percentile calculation}
0:   if value_weights is provided then
0:      $C' \leftarrow [\text{counter}[v_j] \times \text{value\_weights}[\text{item\_id}][v_j] \text{ for } v_j \in V]$ 
0:   else {Uniform binning case}
0:      $C' \leftarrow [\text{counter}[v_j] \text{ for } v_j \in V]$ 
0:   thresholds  $\leftarrow$  a list of size  $B - 1$ 
0:   if  $N \geq B$  then
0:      $C'_{total} \leftarrow \sum C'$ 
0:      $C'_{cumulative} \leftarrow \text{CumulativeSum}(C')$ 
0:     for  $p = 1$  to  $B - 1$  do
0:        $target\_count \leftarrow C'_{total} \times p/B$ 
0:        $idx \leftarrow \text{FindFirstIndexWhere}(C'_{cumulative} > target\_count)$ 
0:       thresholds[ $p - 1$ ]  $\leftarrow V[idx + 1]$  {Handle edge cases}
0:   else {Apply specific assignment for sparse values (centering or striding)}
0:     thresholds  $\leftarrow$  Generate thresholds based on sparse assignment logic
0:   bin_thresholds[item_id]  $\leftarrow$  thresholds
0: return bin_thresholds = 0
    
```

B.2. Dual-Calendar Rotary Position Embedding

To address the unique temporal characteristics of Electronic Health Record (EHR) data—namely, the highly irregular event intervals and the clinical significance of calendrical time—we propose **Dual-Calendar Rotary Position Embedding (RoPE)**. This method extends the conventional RoPE by partitioning the embedding dimension within each attention head to jointly encode both the **relative sequence order** of tokens and their **absolute calendrical time**.

For a given query or key vector $z \in \mathbb{R}^{d_k}$ in an attention head, we partition it into two subspaces: a positional component $z_{pos} \in \mathbb{R}^{d_{pos}}$ and a temporal component $z_{time} \in \mathbb{R}^{d_{time}}$, where $d_k = d_{pos} + d_{time}$.

$$z = [z_{pos} \parallel z_{time}]$$

Each component is then rotated using a specialized RoPE variant before being concatenated back together.

B.2.1. POSITIONAL DIMENSION ENCODING

The z_{pos} component, corresponding to the first d_{pos} dimensions, employs the standard RoPE formulation. With a reduced dimensionality of d_{pos} , this component does not rescale its rotational frequencies to cover a wide positional range. Instead, it effectively **truncates the frequency spectrum**, retaining the high-frequency rotations corresponding to the initial dimensions of a standard RoPE. This strategic choice is predicated on the observation that its primary role is now to disambiguate the order of co-occurring events that share an identical timestamp. For a token at position m , the rotation angle $\theta_{p,i}^{(pos)}$ is defined as:

$$\theta_{p,i}^{(pos)} = \frac{p}{\text{base}^{2i/d}} \quad , \quad i \in \{1, 2, \dots, d_{pos}/2\}$$

The task of modeling long-range temporal dependencies is thus naturally offloaded to the Calendar-Time dimension, which is explicitly designed for this purpose.

B.2.2. CALENDAR-TIME DIMENSION ENCODING

The core novelty of our approach lies in the encoding applied to the z_{time} component. This component is designed to encode an event’s absolute timestamp, t , by capturing its periodicity across multiple, clinically relevant time scales. This is achieved using a predefined set of **semantically meaningful calendrical periods**, $S = \{s_1, s_2, \dots, s_{d_{time}/2}\}$, as detailed in Table 10. These periods capture periodicities ranging from short-term diurnal patterns to long-term annual and multi-year trends.

For each period s_j from this set, we calculate a unique rotation angle $\theta_{t,j}^{(time)}$ that represents the phase of the event within that specific period. The formula for the rotation angle is:

$$\theta_{t,j}^{(time)} = \left(\frac{t \pmod{s_j}}{s_j} \right) \cdot 2\pi$$

This mechanism produces a **multi-scale temporal representation**. Events occurring at the same time of day but on different dates will share the exact same rotation for the ‘day’ period, allowing the model to easily learn periodical patterns.

B.2.3. INTEGRATION AND APPLICATION

Finally, the two rotated components are concatenated to form the final query and key vectors. The full transformation for a query vector $q = [q_{pos} \parallel q_{time}]$ to its rotated form q' is:

$$q' = [\text{RoPE}(q_{pos}, \theta^{(pos)}) \parallel \text{RoPE}(q_{time}, \theta^{(time)})]$$

An identical transformation is applied to the key vector k . By equipping the self-attention mechanism with this dual-encoding strategy, our model can simultaneously reason about the sequential flow of information and the absolute, cyclical context of clinical events.

Table 10. Predefined Calendrical Periods for Temporal Encoding

Category	Period Name	Duration (seconds)
<i>Short-term</i>	5 minutes	300
	10 minutes	600
	30 minutes	1,800
	1 hour	3,600
	3 hours	10,800
	12 hours	43,200
<i>Mid-term</i>	1 day	86,400
	2 days	172,800
	1 week	604,800
	2 weeks	1,209,600
	1 month	2,629,746
	1 season (3 months)	7,889,238
	6 months	15,778,476
<i>Long-term</i>	1 year	31,556,952
	2 years	63,113,904
	4 years	126,227,808
	10 years	315,569,520
	30 years	946,708,560
	100 years	3,155,695,200
	300 years	9,467,085,600

B.3. Time Conditioned Foresee Objective

To elucidate the mechanics of our proposed Time-Conditioned Foresee (TCF) module, we provide a step-by-step explanation. This description follows the flow of information from the initial input—the final hidden state of a backbone model—to the module’s dual outputs. The detailed computational flow is presented in Algorithm 3.

We define the primary dimensions:

- B denotes the batch size.
- L denotes the sequence length.
- d_{model} represents the hidden dimension of the backbone model’s output.
- d_{embed} is the dimensionality of our internal time embeddings, set to 32.
- N_{scales} is the number of time scales used for decomposition, 11 in our implementation.
- N_{foresee} is the number of future timestamps provided for the foresee objective, set to 10.
- C_i is the number of discrete categories for the i -th time scale.

The process begins with the final hidden state from the backbone decoder for each token in the sequence.

Input: The last hidden state tensor, H_{last} .

- $H_{\text{last}} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$

B.3.1. A. NEXT EVENT TIME PREDICTION PATH ($\mathcal{L}_{\text{NEXT_TIME}}$)

This pathway is responsible for predicting the time until the next event.

1. Initial Projection The input hidden state H_{last} is passed through a two-layer Feed-Forward Network (FFN), denoted as FFN_{enc} , to create a representation for time prediction.

- **Input:** $H_{\text{last}} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$
- **Output:** An intermediate time-focused tensor, H_{time} .
 - $H_{\text{time}} = \text{FFN}_{\text{enc}}(H_{\text{last}}) \in \mathbb{R}^{B \times L \times (N_{\text{scales}} \cdot d_{\text{embed}})}$

2. Logit Generation The tensor H_{time} is conceptually partitioned into N_{scales} segments. Each segment is used to compute the logits for its corresponding time scale by multiplying it with the respective time embedding weight matrix.

- **Input:** H_{time} , treated as N_{scales} tensors $\{H_{\text{time}}^{(i)}\}_{i=1}^{N_{\text{scales}}}$, where each $H_{\text{time}}^{(i)} \in \mathbb{R}^{B \times L \times d_{\text{embed}}}$.
- **Operation:** For each scale i , we compute logits: $\text{Logits}^{(i)} = H_{\text{time}}^{(i)} \cdot (W_{\text{embed}}^{(i)})^T$, where $W_{\text{embed}}^{(i)} \in \mathbb{R}^{C_i \times d_{\text{embed}}}$.
- **Output:** A set of N_{scales} logit tensors.
 - $\text{Logits}^{(i)} \in \mathbb{R}^{B \times L \times C_i}$

3. Ground-Truth Label Decomposition To compute the loss, these logits are compared against ground-truth labels. Instead of regressing a continuous time value, we transform the ground-truth time delta (in seconds) into a set of categorical integer labels. This is achieved through a deterministic process analogous to a mixed-radix conversion, using the time scales defined in Table 11.

For instance, assume a ground-truth time delta ΔT_{next} of **34,586,130 seconds**. The conversion to a vector of N_{scales} integer labels proceeds sequentially from the largest time scale to the smallest, using integer division to find the label and the modulo operator to find the remainder for the next step.

1. **year10:** $34,586,130 // 315,360,000 = 0$. Remainder: 34,586,130. → **Label: 0**
2. **year1:** $34,586,130 // 31,536,000 = 1$. Remainder: 3,050,130. → **Label: 1**

Table 11. Time scales for multi-scale decomposition, along with their duration in seconds and the number of categories for classification.

Time Scale	Duration in Seconds	Num. of Categories
year10	315,360,000	10
year1	31,536,000	10
month3	7,948,800	4
month1	2,678,400	3
week1	604,800	5
day1	86,400	7
hour6	21,600	4
hour1	3,600	6
minute10	600	6
minute1	60	10

3. **month3**: $3,050,130 // 7,948,800 = 0$. Remainder: 3,050,130. → **Label: 0**
4. **month1**: $3,050,130 // 2,678,400 = 1$. Remainder: 371,730. → **Label: 1**
5. **week1**: $371,730 // 604,800 = 0$. Remainder: 371,730. → **Label: 0**
6. **day1**: $371,730 // 86,400 = 4$. Remainder: 27,330. → **Label: 4**
7. **hour6**: $27,330 // 21,600 = 1$. Remainder: 5,730. → **Label: 1**
8. **hour1**: $5,730 // 3,600 = 1$. Remainder: 2,130. → **Label: 1**
9. **minute10**: $2,130 // 600 = 3$. Remainder: 330. → **Label: 3**
10. **minute1**: $330 // 60 = 5$. Remainder: 30. → **Label: 5**
11. **position**: A mechanism to account for events occurring simultaneously at the same time. For the next event label, it is set to 0; for subsequent foresee labels, it is set to +1 if the time is the same as the previous one, and 0 otherwise. → **Label: 0**

Ultimately, the continuous value of 34,586,130 seconds is converted into the following vector of ten integer labels, which constitutes the ground-truth $Y_{\text{next.time}}^{(i)}$ for this example:

$$[0, 1, 0, 1, 0, 4, 1, 1, 3, 5, 0]$$

By training the model to predict these categorical labels for each time scale, we transform a difficult regression task into a series of more stable and effective classification tasks. The final scalar loss, $\mathcal{L}_{\text{next.time}}$, is the average Cross-Entropy loss calculated between the generated logits and these decomposed ground-truth labels.

B.3.2. B. TIME-CONDITIONING PATH FOR FORESEE OBJECTIVE

This pathway conditions the hidden state on a set of specified future timestamps to predict upcoming events.

1. Input Foresee Timestamps The module receives future time deltas from the N_{foresee} future events relative to the current timestamp at each position.

- **Input:** A tensor of future time deltas, $\Delta T_{\text{foresee}} \in \mathbb{Z}^{B \times L \times N_{\text{foresee}}}$.

2. Time Embedding Each time delta in $\Delta T_{\text{foresee}}$ is decomposed into N_{scales} integer labels (as demonstrated above). These labels are used to look up corresponding vectors from the embedding tables, $\{W_{\text{embed}}^{(i)}\}_{i=1}^{N_{\text{scales}}}$, which are then concatenated. *Note, we share the weights for embedding and unembedding timestamps.

- **Input:** Decomposed time labels, $C_{\text{foresee}} \in \mathbb{Z}^{B \times L \times N_{\text{foresee}} \times N_{\text{scales}}}$.
- **Output:** A dense time embedding tensor, E_{time} .
 - $E_{\text{time}} \in \mathbb{R}^{B \times L \times N_{\text{foresee}} \times (N_{\text{scales}} \cdot d_{\text{embed}})}$

3. Projection of Time Embedding The concatenated embedding E_{time} is projected to the model’s hidden dimension via FFN_{dec} .

- **Input:** $E_{\text{time}} \in \mathbb{R}^{B \times L \times N_{\text{foresee}} \times (N_{\text{scales}} \cdot d_{\text{embed}})}$
- **Output:** A processed time conditioning tensor, $H_{\text{time_cond}}$.

$$- H_{\text{time_cond}} = \text{FFN}_{\text{dec}}(E_{\text{time}}) \in \mathbb{R}^{B \times L \times N_{\text{foresee}} \times d_{\text{model}}}$$

4. Time-Conditioning via Fusion The final step fuses the original hidden state H_{last} with the processed time conditioning tensor $H_{\text{time_cond}}$. To align their dimensions for the element-wise addition, H_{last} is first expanded by inserting a new dimension. This prepares it for broadcasting across the N_{foresee} dimension, allowing each of the N_{foresee} time embeddings to condition the single original hidden state.

- **Inputs:**
 - Original hidden state: $H_{\text{last}} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$
 - Time conditioning tensor: $H_{\text{time_cond}} \in \mathbb{R}^{B \times L \times N_{\text{foresee}} \times d_{\text{model}}}$
- **Operation:** The fusion is performed via a residual connection. First, H_{last} is unsqueezed, and then added to $H_{\text{time_cond}}$.
 - $H'_{\text{last}} = \text{Unsqueeze}(H_{\text{last}}, \text{dim} = 2) \in \mathbb{R}^{B \times L \times 1 \times d_{\text{model}}}$
 - $H_{\text{fused}} = H'_{\text{last}} + H_{\text{time_cond}}$ // Broadcasting occurs along the N_{foresee} dimension.
- **Output:** The final time-conditioned hidden state, $H_{\text{conditioned}}$, after Layer Normalization and a final FFN block.
 - $H_{\text{conditioned}} \in \mathbb{R}^{B \times L \times N_{\text{foresee}} \times d_{\text{model}}}$

SUMMARY OF MODULE OUTPUTS

The TCF module produces two primary outputs:

1. **Next Time Loss ($\mathcal{L}_{\text{next_time}}$):** A scalar value for backpropagation.
2. **Conditioned Hidden State ($H_{\text{conditioned}}$):** A tensor of shape $\mathbb{R}^{B \times L \times N_{\text{foresee}} \times d_{\text{model}}}$, which serves as the input to the final prediction head for calculating the foresee loss, $\mathcal{L}_{\text{foresee}}$. The ground-truth label corresponding to each conditioned hidden vector is the `Feature` token of the actual clinical event that occurred at the given foresee timestamp.

Algorithm 3 Time-Conditioned Foresee (TCF) Module

Require:

- 0: $H_{\text{last}} \in \mathbb{R}^{B \times L \times d_{\text{model}}}$: Last hidden states from the backbone model.
- 0: $T_{\text{current}} \in \mathbb{R}^{B \times L}$: Absolute timestamps for each hidden state in H_{last} .
- 0: $T_{\text{next}} \in \mathbb{R}^{B \times L}$: Absolute timestamps of the next event for each position.
- 0: $T_{\text{foresee}} \in \mathbb{R}^{B \times L \times N_{\text{foresee}}}$: A set of absolute future timestamps for conditioning.
- 0: $\mathcal{W}$: All trainable weights, including FFNs and embedding tables $\{W_{\text{embed}}^{(i)}\}_{i=1}^{N_{\text{scales}}}$.
- 0: Periods : A dictionary mapping each time scale to its duration in seconds.

Ensure:

- 0: $\mathcal{L}_{\text{next_time}} \in \mathbb{R}$: The loss for next event time prediction.
- 0: $H_{\text{conditioned}} \in \mathbb{R}^{B \times L \times N_{\text{foresee}} \times d_{\text{model}}}$: Hidden states conditioned on T_{foresee} .

```

0: function TCF_MODULE( $H_{\text{last}}, T_{\text{current}}, T_{\text{next}}, T_{\text{foresee}}, \mathcal{W}, \text{Periods}$ )
  {Part A: Next Event Time Prediction}
  0:  $\Delta T_{\text{next}} \leftarrow T_{\text{next}} - T_{\text{current}}$ 
  0:  $Y_{\text{next}} \leftarrow \text{DECOMPOSETIME}(\Delta T_{\text{next}}, \text{Periods})$ 
  0:  $H_{\text{time}} \leftarrow \text{FFN}_{\text{enc}}(H_{\text{last}})$  {Shape:  $(B, L, N_{\text{scales}} \cdot d_{\text{embed}})$ }
  0:  $H_{\text{time}} \leftarrow \text{RESHAPE}(H_{\text{time}}, (B, L, N_{\text{scales}}, d_{\text{embed}}))$ 
  0:  $\mathcal{L}_{\text{total}} \leftarrow 0$ 
  0: for  $i = 1 \rightarrow N_{\text{scales}}$  do
  0:    $H_{\text{time}}^{(i)} \leftarrow H_{\text{time}}[:, :, i, :]$  {Shape:  $(B, L, d_{\text{embed}})$ }
  0:    $\text{Logits}^{(i)} \leftarrow H_{\text{time}}^{(i)} \cdot (W_{\text{embed}}^{(i)})^T$ 
  0:    $Y_{\text{next}}^{(i)} \leftarrow Y_{\text{next}}[:, :, i]$ 
  0:    $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{total}} + \text{CROSSENTROPYLOSS}(\text{Logits}^{(i)}, Y_{\text{next}}^{(i)})$ 
  0:  $\mathcal{L}_{\text{next\_time}} \leftarrow \mathcal{L}_{\text{total}} / N_{\text{scales}}$ 
  {Part B: Time-Conditioning for Foresee Objective}
  0:  $T'_{\text{current}} \leftarrow \text{UNSQUEEZE}(T_{\text{current}}, \text{dim} = 2)$  {Shape:  $(B, L, 1)$ }
  0:  $\Delta T_{\text{foresee}} \leftarrow T_{\text{foresee}} - T'_{\text{current}}$ 
  0:  $C_{\text{foresee}} \leftarrow \text{DECOMPOSETIME}(\Delta T_{\text{foresee}}, \text{Periods})$  {Shape:  $(B, L, N_{\text{foresee}}, N_{\text{scales}})$ }
  0:  $E_{\text{time\_list}} \leftarrow []$ 
  0: for  $i = 1 \rightarrow N_{\text{scales}}$  do
  0:    $C_{\text{foresee}}^{(i)} \leftarrow C_{\text{foresee}}[:, :, :, i]$ 
  0:    $E_{\text{time}}^{(i)} \leftarrow \text{LOOKUP}(W_{\text{embed}}^{(i)}, C_{\text{foresee}}^{(i)})$  {Shape:  $(B, L, N_{\text{foresee}}, d_{\text{embed}})$ }
  0:   Append  $E_{\text{time}}^{(i)}$  to  $E_{\text{time\_list}}$ 
  0:  $E_{\text{time}} \leftarrow \text{CONCATENATE}(E_{\text{time\_list}}, \text{dim} = -1)$  {Shape:  $(B, L, N_{\text{foresee}}, N_{\text{scales}} \cdot d_{\text{embed}})$ }
  0:  $H_{\text{time\_cond}} \leftarrow \text{FFN}_{\text{dec}}(E_{\text{time}})$  {Shape:  $(B, L, N_{\text{foresee}}, d_{\text{model}})$ }
  0:  $H'_{\text{last}} \leftarrow \text{UNSQUEEZE}(H_{\text{last}}, \text{dim} = 2)$  {Shape:  $(B, L, 1, d_{\text{model}})$ }
  0:  $H_{\text{fused}} \leftarrow \text{LAYERNORM}(H'_{\text{last}} + H_{\text{time\_cond}})$ 
  0:  $H_{\text{conditioned}} \leftarrow \text{LAYERNORM}(H_{\text{fused}} + \text{FFN}_{\text{final}}(H_{\text{fused}}))$ 
  0: return  $\mathcal{L}_{\text{next\_time}}, H_{\text{conditioned}}$ 

0: function DECOMPOSETIME( $\Delta T, \text{Periods}$ ) {Helper function for time decomposition}
0:    $R \leftarrow \Delta T$ 
0:    $\text{Labels} \leftarrow []$ 
0:   for scale in REVERSED( $\text{Periods.keys()}$ ) do
0:      $L_{\text{scale}} \leftarrow R // \text{Periods}[\text{scale}]$  {Integer division}
0:      $R \leftarrow R \% \text{Periods}[\text{scale}]$  {Modulo operation}
0:      $L_{\text{scale}} \leftarrow \text{CLAMP}(L_{\text{scale}}, \text{min} = 0, \text{max} = C_{\text{scale}} - 1)$ 
0:     Prepend  $L_{\text{scale}}$  to  $\text{Labels}$ 
0:   return STACK( $\text{Labels}$ )
=0
    
```

C. Data and Preprocessing

C.1. Dataset: MIMIC-III

MIMIC-III (Johnson et al., 2016a;b) is an openly accessible resource that contains de-identified clinical data from more than 40,000 individuals admitted to the intensive care units of Beth Israel Deaconess Medical Center between 2001 and 2012. The dataset encompasses a wide range of information, including patient demographics, hourly vital sign recordings, laboratory measurements, administered treatments and procedures, prescribed medications, clinical notes, radiology reports, and outcomes such as in-hospital and post-discharge mortality.

The MIMIC-III database was de-identified in compliance with Health Insurance Portability and Accountability Act (HIPAA) standards through data cleansing and systematic date shifting. To preserve clinical intervals, patient-specific dates were consistently shifted into the future by a random offset, placing admissions within the years 2100–2200 while retaining the original time of day, weekday, and approximate seasonality. For patients older than 89, dates of birth were modified such that their recorded ages exceed 300 years, thereby masking their true age in accordance with HIPAA requirements. This modification provides a suitable framework for our Dual-Calendar RoPE, which is designed to address calendrical time, to operate effectively.

C.1.1. FEATURE SELECTION

Electronic Health Records (EHRs) are rich with events that have a numerical `Value`, a characteristic that distinguishes them from natural language. Consequently, our experiments focused on events that possess a numerical `Value`. We utilized 17 vital sign features from the `CHARTEVENTS.csv` file, following the selection in (Harutyunyan et al., 2019), and the top 100 most frequently measured laboratory tests from `LABEVENTS.csv` as the events for our study. These features are detailed in Table 12. In addition, patient events such as “hospital admission”, “ICU transfer (ICU in)”, “ICU discharge (ICU out)”, and “hospital discharge” were utilized.

C.1.2. FURTHER PREPROCESSING

In addition to the preprocessing of (Harutyunyan et al., 2019), we made the following modifications: (1) removed outlier values in the laboratory data based on independent evaluations by three physicians and standardized the measurement units; (2) excluded hospitalization episodes with fewer than 10 events; and (3) added an anchor token with a timestamp of January 1st, 00:00 of the same year before each admission token to serve as a calendrical time reference. As a result, the lengths of patients’ medical histories follow the distribution shown in Figure 6.

C.2. Dataset: MIMIC-IV

MIMIC-IV (Johnson et al., 2023) is a comprehensive, de-identified EHR dataset containing clinical data for over 364,000 unique patients admitted to the Beth Israel Deaconess Medical Center between 2008 and 2022. It features a modular architecture providing a rich longitudinal view that includes demographics, laboratory measurements, medications, and emergency department observations.

To ensure HIPAA compliance, patient-specific dates are shifted into the future by a random offset, placing records within the years 2100–2200 while strictly preserving clinical time intervals and seasonality. Notably, for patients older than 89, ages are fixed at 91 to prevent re-identification. This systematic temporal organization provides a robust framework for time-series modeling and evaluation of calendrical time-aware architectures, such as our Dual-Calendar RoPE.

C.2.1. FEATURE SELECTION

We utilized 17 vital sign features from the `CHARTEVENTS.csv` file, and the top 100 most frequently measured laboratory tests from `LABEVENTS.csv` as the events for our study. These features are detailed in Table 12. In addition, patient events such as “hospital admission”, “ICU transfer (ICU in)”, “ICU discharge (ICU out)”, and “hospital discharge” were utilized.

Table 12. Selected features from MIMIC-III used in this study.

Category	Features
Ethnicity - # 10 (from <code>ADMISSIONS.csv</code>)	White, White - Russian, White - other European, Asian, Asian - Chinese, Hispanic or Latino, Hispanic/Latino - Dominican, Black/Cape Verdean, Black/African American, Others or Unknown
Vital Signs - # 17 (from <code>CHARTEVENTS.csv</code>)	Capillary refill rate (CRR), Systolic blood pressure (SBP), Mean blood pressure (MBP), Diastolic blood pressure (DBP), Fraction of inspired oxygen (FiO2), Heart rate (HR), Respiratory rate (RR), Glasgow coma scale eye response (GCS-E), Glasgow coma scale motor response (GCS-M), Glasgow coma scale verbal response (GCS-V), Glasgow coma scale (GCS), Serum glucose, O2 saturation, Blood pH, Body temperature, Height, Weight.
Laboratory Tests - # 100 (from <code>LABEVENTS.csv</code>)	Hematocrit, Potassium, Sodium, Creatinine, Chloride, Blood urea nitrogen, Bicarbonate, Platelets, Anion gap, White blood cell count, Hemoglobin chemistry, Mean corpuscular hemoglobin concentration, Red blood cell count, Mean corpuscular hemoglobin, Mean corpuscular volume, Red Cell Distribution Width, Magnesium, Calcium Total, Phosphate, Base excess, CO2 (ETCO2, PCO2, etc.), Partial pressure of oxygen, Partial pressure of carbon dioxide, Partial thromboplastin time, Prothrombin time INR, Prothrombin time, Calcium Free, Bilirubin Total, Alanine aminotransferase, Aspartate aminotransferase, Alkaline phosphate, Potassium blood gas, Lactate, Lymphocytes, Neutrophils, Monocytes, Eosinophils, Basophils, Albumin, Creatine Kinase, Oxygen blood gas, Urine Specific Gravity, Creatine Kinase-MB, Lactate dehydrogenase, Urine Protein, Urine Urobilinogen, Urine Ketone, Urine Color, Urine Appearance, Urine Blood, Urine Bilirubin, Urine Nitrite, Urine Leukocyte, Hematocrit blood gas, Hemoglobin blood gas, Troponin-T, Positive end-expiratory pressure, Urine Yeast, Urine White blood cell count, Urine Red blood cell count, Urine Epithelial cells, Band Neutrophils, Urine Bacteria, Sodium blood gas, Lipase, Amylase, Estimated GFR, Hypochromia, Anisocytosis, Macrocytosis, Lymphocytes Atypical, Metamyelocytes, Myelocytes, Microcytes, Poikilocytosis, Vancomycin (blood), Chloride blood gas, Polychromasia, Functional Fibrinogen, Bilirubin Direct, Bilirubin Indirect, Platelet Smear, Urine Creatinine, Thyroid-stimulating hormone, Urine Sodium, Triglycerides, Granulocyte count, CK-MB Index, Phenytoin (blood), Alveolar-arterial gradient, Cholesterol Total, Urine osmolality, Osmolality, Uric acid, Cholesterol HDL, Iron, Cholesterol ratio Total/HDL, Ferritin, Transferrin, Iron binding capacity, HbA1C, Nucleated red cells, Cholesterol, Ovalocyte, Urine Hyaline casts, Urine mucous, Cortisol, Urine urea nitrogen, Haptoglobin, Protein (Total), Vitamin B12, Benzodiazepine Screen, Barbiturate Screen, Tricyclic Antidepressant Screen, Troponin-I, Urine potassium, Tacrolimus level, Schistocytes, Reticulocyte count, Ethanol, Urine Chloride, Acetaminophen, Urine Cocaine, Urine Benzodiazepine screen, Urine Amphetamine screen, Urine Opiate screen, Urine Barbiturate screen, Urine Methadone, Bicarbonate blood gas, Salicylate, Urine Total protein, Teardrop cells, Cyclosporin, Folate, Burr cells, Sedimentation rate, Digoxin, Thyroxine, Globulin, Urine protein/creatinine ratio, NT-proBNP, Urine Amorphous crystal, C-reactive protein, Large platelets, Urine Granular casts, Gentamicin, Target cell, Transitional epithelial cells, Fibrin degradation, CSF Lymphs.

D. Backbone, Baselines, Pre-training detail

D.1. Backbone architecture

The backbone model used in this study follows a standard **decoder-only transformer** architecture. To minimize performance variations caused by differences in backbone models and to quantitatively assess the effectiveness of our proposed training methodology, we used the same backbone across all experiments. However, for models trained with Transformer encoders using the masked language modeling approach (HEART, TRADE), we removed the causal mask during pre-training so that

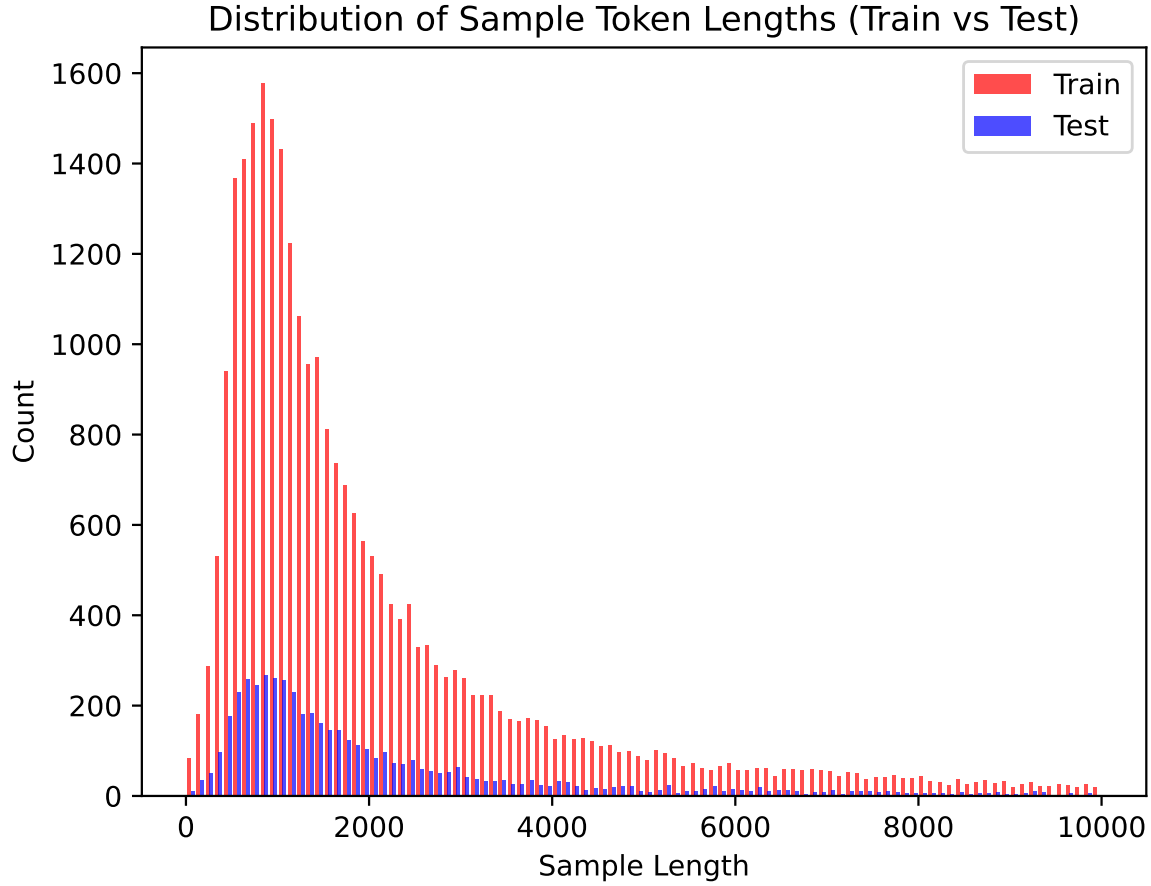


Figure 6. Length distribution of tokenized histories per patient. Most are within 2048 tokens. Lengths exceeding 10k tokens are not shown.

they could be used as encoder models. The backbone details are as follows:

- **Vocabulary Size:** 155 (1208 if not share bin)
- **Embedding and Hidden Dimension** (d_{model}): 512
- **Number of Decoder Layers** (N): 6
- **Number of Attention Heads:** 8
- **Dimension per Head:** 64
- **Dimension of K, Q:** 64 (Ours: first 24: positional RoPE / last 40: calendrical time RoPE)
- **Feed-Forward Network (FFN) Inner Dimension** (d_{ff}): 2048
- **Total Parameters (Backbone):** 19,001,344 (19,060,736 if not share bin)
- **Activation:** ReLU (Agarap, 2018)

Each decoder layer is composed of two main sub-layers: a **multi-head self-attention** block and a **feed-forward network**. For training stability, the model adopts a **Pre-Layer Normalization (Pre-LN)** structure, where Layer Normalization is applied to the input of each sub-layer. A **residual connection** is then employed around each of the two sub-layers.

D.2. Baseline models

We compared our model with baselines that utilized either Time or Value in their training: HEART, FM4EHR, MOTOR, STraTS, EHRSHOT, TRADE, EHRMamba, and ETHOS. The backbone for each model was standardized as described above, while other methodologies (binning, tokenization, positional embedding, learning objective, etc.) followed their original papers. The reproduction details are as follows.

HEART (Total Parameters: 20,204,180)

- **Backbone:** Uses a Transformer encoder due to its MLM-based loss.
- **Bin Sharing:** No, uses Value-Feature paired tokenization.
- **Binning:** 10-uniform binning.
- **Positional Embedding:** Absolute positional embedding—a learned positional embedding of the visit index (0 for the patient’s first visit, 1 for the second, and so on).
- **Learning Objective:** For each visit, event tokens (Value-Feature paired) are masked with a probability of $p_{\text{mask}} = 0.15$. The model is trained via a multi-class classification loss to predict the masked token. This is trained separately for different event types (V/S, Lab). Additionally, for unmasked events, values are altered with a probability of $p_{\text{anomaly}} = 0.05$, and the model is trained with a binary classification loss to identify whether the value at each position was altered.

FM4EHR (Total Parameters: 19,001,536)

- **Backbone:** Transformer decoder.
- **Bin Sharing:** Yes, Feature and Value are tokenized separately. Numeric values share bin tokens across different features.
- **Binning:** 10-uniform binning.
- **Positional Embedding:** Uses rotary positional embedding based only on position, without explicit time information.
- **Learning Objective:** Uses NTP loss.

MOTOR (Total Parameters: 20,341,952)

- **Backbone:** Transformer decoder.
- **Methodology Adaptation:** This model was originally designed for feature-only events (e.g., diagnosis codes). To adapt it for continuous values, we set the “occurrence of an abnormal measurement result” as the endpoint for its time-to-event loss. An abnormal value was defined by either being outside the medical normal range (per Harrison’s Principles of Internal Medicine (Kasper et al., 2015), evaluated by three clinicians) or being a statistical outlier (e.g., top/bottom 5%).
- **Bin Sharing:** No, uses Value-Feature paired tokenization.
- **Binning:** 10-uniform binning was applied to the Value.
- **Positional Embedding:** The model converts each event’s timestamp into ‘days since birth’ and applies this value in its rotary positional embedding.
- **Learning Objective:** Uses the Time-to-event loss from the original paper.

STraTS (Total Parameters: 19,374,801)

- **Architecture:** As this is an older paper, its structure is not suitable for parallel training. We therefore modified the architecture while retaining the core ideas. We used the last hidden state of the decoder at each time step as the event embedding, instead of the original Fusion Self-Attention mechanism.
- **Value and Time Embedding:** This model does not use binning. Instead, it embeds Features via a look-up table and continuously embeds Value and Time (in hours) via a 2-layer fully connected layer. The resulting embeddings are summed to form the final event embedding.
- **Positional Embedding:** No additional positional embedding is used beyond the Time information included in the event embedding.
- **Demographics:** Unlike typical models, STraTS encodes demographic information (gender, race, age) with a separate MLP and concatenates it to the last hidden state.
- **Learning Objective:** The final embedding is used to predict the value of events occurring within two hours of each event, trained with a mean squared error loss.

EHRSHOT (Total Parameters: 20,278,892)

- **Backbone:** Transformer decoder.
- **Bin Sharing:** No, uses Value-Feature paired tokenization.
- **Binning:** 10-uniform binning.
- **Positional Embedding:** Following CLMBR-T (Steinberg et al., 2021), it uses Rotary positional embedding based on position order. Time information is not used beyond ordering for this component.
- **Time Information:** A 5-dimensional time vector is concatenated to the token embedding vector. This vector consists of the z-normalized values of [age, log(age), days since admission, log(days since admission), first admission indicator]. The final concatenated vector length is 512.
- **Learning Objective:** Next token prediction modeling.

TRADE (Total Parameters: 21,617,664)

- **Backbone:** Uses a Transformer encoder due to its MLM-based loss.
- **Bin Sharing:** No, uses Value-Feature paired tokenization.
- **Binning:** 9-standard deviation-based binning. For each feature’s value distribution, thresholds are set by adding $\{-10, -3, -1, -0.5, 0.5, 1, 3, 10\}$ standard deviations to the mean, creating 9 bins. This method can be sensitive to outliers, so additional clipping was performed on 17 vital sign data points based on physician guidelines. The clipping ranges are as follows: CRR:[0,1], SBP:[0,400], MBP:[0,300], DBP:[0,300], FiO2:[0,1], HR:[0,200], RR:[0,100], GCS-E:[1,4], GCS-M:[1,6], GCS-V:[1,5], Glucose:[0,1200], O2 saturation:[0,100], Body temperature:[20, 45], Height:[0,1000], Weight:[0,1000].
- **Positional Embedding:** Absolute positional embedding—performs learned positional embedding using three types of sequential and temporal information: 1) The index of the current hospital admission, 2) The number of days passed since admission, and 3) The current age. These are all integers, passed through an embedding layer, and then summed.
- **Learning Objective:** Uses a standard MLM methodology, masking each token with $p = 0.2$ and using the last hidden state of the masked position to predict the pre-mask label via a classification loss.

EHRMamba (Total Parameters: 21,770,752)

- **Backbone:** Although the original paper uses Mamba, we applied the same Transformer decoder backbone for a fair comparison.
- **Bin Sharing:** No, uses Value-Feature paired tokenization.
- **Binning:** 10-uniform binning (the original paper used 5-uniform binning, but we matched the bin count for a fair comparison).
- **Positional Embedding:** Uses four types of absolute positional embeddings, which are summed: (1) Learned PE based on the hospital visit number, (2) Learned PE based on token type, (3) Time embedding based on age using the Time2Vec model (Kazemi et al., 2019), and (4) Position-based sin/cos positional embedding from (Vaswani et al., 2017).
- **Learning Objective:** Uses a next token prediction loss.

ETHOS (Total Parameters: 20,050,944)

- **Backbone:** Transformer decoder.
- **Bin Sharing:** Yes, Feature and Value are tokenized separately. Numeric values share bin tokens across different features.
- **Binning:** 10-uniform binning.
- **Time Information:** Following the original paper, the time interval between each event is converted into one of 13 discrete tokens, which are inserted into the sequence between event tokens.
- **Positional Embedding:** Uses a learned positional embedding based on position.
- **Learning Objective:** Uses NTP loss.

D.3. Pre-training Details

Pre-training was conducted using the training dataset. The hyperparameters were fixed as follows: the batch size was 64, and the number of training epochs was 50. We used the Adam (Kingma & Ba, 2014) optimizer with a learning rates of $\{5 \times 10^{-4}, 1 \times 10^{-4}, 5 \times 10^{-5}, 1 \times 10^{-5}\}$. A 50-step warmup was employed, followed by a cosine annealing schedule that reduced the learning rate to 1/100 of its initial value. Gradient clipping was applied with a threshold of 1.0. The training was performed using either 4 NVIDIA A40 GPUs or 2 NVIDIA RTX PRO 6000 Blackwell GPUs, with Distributed Data Parallel training at a per-GPU batch size of 16 or 32, respectively. The implementation was based on Python version 3.12 and PyTorch (Paszke et al., 2017) version 2.8.0.

E. Downstream tasks and Fine-tuning**E.1. Downstream tasks**

Existing EHR foundation models generally lack generative capabilities, and thus evaluating performance on diverse clinical downstream tasks has been a common practice (Fallahpour et al., 2025; Huang et al., 2024; Burkhart et al., 2025; Renc et al., 2024). While our model possesses strong generative properties, we follow the line of prior work and perform downstream task evaluations to assess the quality of patient representations at each timestamp. We first adopted the four widely used downstream tasks from the MIMIC-III benchmark (Harutyunyan et al., 2019): In-hospital Mortality (IHM), Decompensation-death (Dec-D), Length of Stay (LOS), and Phenotyping (Phe). For these tasks, labels were obtained following the original preprocessing pipelines. To further examine whether the model can capture patient states beyond simple deterioration, we added three additional tasks: Decompensation-arrest (Dec-A), Oliguria/Anuria (HUO), and Vasopressor (Vaso) use. Dec-A is similar to Dec-D, but includes arrest events from `CHARTEVENTS.csv` in addition to death as decompensation events; the task aims to predict deterioration 24 hours in advance. HUO labels indicate whether the patient currently exhibits oliguria or anuria. Specifically, oliguria is defined as urine output below 0.5 mL/kg/hr for at least 6 hours, and anuria as

below 0.1 mL/kg/hr for at least 6 hours. Labels were set for patients with available weight and hourly urine output data, while patients with incomplete information were excluded from training. Vaso labels indicate whether a patient is currently receiving vasopressors. Positive labels were assigned if administration records for Vasopressin, Dobutamine, Epinephrine, Norepinephrine, or Dopamine were present in `INPUTEVENTS_CV.csv` or `INPUTEVENTS_MV.csv`; negative labels were assigned if other medications were administered but no vasopressors were recorded.

All downstream tasks were measured at the hospitalization level following the original papers (Harutyunyan et al., 2019) (i.e., if a patient had multiple hospital admissions, each admission was treated as a separate EHR sequence). Table 13 provides detailed statistics, including the number of hospitalizations available for each task and the label distribution for each class.

Table 13. Prediction time and label counts for each downstream task. For binary classification, the positive/negative counts are shown; for multiclass classification, the counts for each class are shown. For tasks with multiple sub-tasks, each sub-task’s label counts are shown.

Task	Prediction point	# Hospitalization	# Sub task & # Class	Train labels	Test labels
IHM	48h after ICU adm.	17,903 / 3,236	1 / 2	2424 / 15479	374 / 2,862
Dec-D	Hourly	35,365 / 6,237	1 / 2	61,018 / 2,847,424	9,684 / 513,552
Dec-A	Hourly	35,365 / 6,237	1 / 2	65,298 / 2,843,144	10,239 / 512,997
LOS	Hourly	35,523 / 6,265	1 / 10	[cls1: 790,196, cls2: 503,423, cls3: 316,774, cls4: 215,075, cls5: 158,987, cls6: 124,146, cls7: 100,890, cls8: 84,241, cls9: 312,111, cls10: 319,619] t1: 7,644 / 28,029 t2: 2660 / 33013 t3: 3657 / 32016 t4: 11434 / 24239 t5: 4821 / 30852 t6: 4675 / 30998 t7: 7349 / 28324 t8: 2565 / 33108 t9: 9550 / 26123 t10: 11497 / 24176 t11: 3444 / 32229 t12: 6869 / 28804	[cls1: 139,682, cls2: 90,478, cls3: 56,289, cls4: 38,795, cls5: 28,542, cls6: 22,225, cls7: 18,077, cls8: 15,145, cls9: 55,997, cls10: 60,710] t1: 374 / 2862 t2: 1331 / 4945 t3: 415 / 5861 t4: 675 / 5601 t5: 2028 / 4248 t6: 831 / 5445 t7: 789 / 5487 t8: 1337 / 4939 t9: 442 / 5834 t10: 1683 / 4593 t11: 2074 / 4202 t12: 593 / 5683
Phe	End of stay	35,563 / 6,273	25 / 2	t13: 10362 / 25311 t14: 14922 / 20751 t15: 9617 / 26056 t16: 2573 / 33100 t17: 4775 / 30898 t18: 3170 / 32503 t19: 1816 / 33857 t20: 1435 / 34238 t21: 3080 / 32593 t22: 4970 / 30703 t23: 6468 / 29205 t24: 5118 / 30555 t25: 2779 / 32894	t13: 1205 / 5071 t14: 1813 / 4463 t15: 2653 / 3623 t16: 1667 / 4609 t17: 495 / 5781 t18: 819 / 5457 t19: 556 / 5720 t20: 355 / 5921 t21: 272 / 6004 t22: 570 / 5706 t23: 852 / 5424 t24: 1111 / 5165 t25: 874 / 5402
HUO	Hourly	11,891 / 2,187	2 / 2	t1: 148737 / 891153 t2: 148737 / 1053830	t1: 56102 / 162429 t2: 26682 / 191849
Vaso	Hourly	35,438 / 6,249	1 / 2	202,765 / 2,747,736	36,306 / 494,215

E.2. Fine-tuning details

The downstream tasks were trained by attaching a task-specific prediction head to the last hidden state of the backbone, applied uniformly to our model and all baselines. Since labels must be inferred using only information available up to each timestep, a causal mask was employed. We froze the pretrained model and trained the prediction heads simultaneously. To evaluate generalization performance under varying input distributions (e.g., absence of laboratory data), we experimented with three settings: (i) the full set of 117 variables (17 vital signs + 100 laboratory measurements), (ii) only 17 vital signs, and (iii) a reduced set of 6 vital signs (SBP, DBP, body temperature, heart rate, respiratory rate, SpO₂). For training, the original training set was split into train/validation subsets with an 85:15 ratio; the train subset was used for optimization, while the validation subset was used for early stopping and hyperparameter search. As in pre-training, the learning rate lr

Table 14. Meta Information

Method	Value sharing	Downstream # features	test-loss	valid-loss	train-loss	Pre-training LR	Downstream LR
HEART	X	117	5.3043	5.3681	5.3714	0.0005	0.0005
HEART	X	17	5.4340	5.4604	5.4713	0.0005	0.0005
HEART	X	6	5.8346	5.8703	5.8702	0.0005	0.0005
MOTOR	X	117	4.9454	4.9020	4.9118	0.0001	0.0005
MOTOR	X	17	5.2117	5.1570	5.1803	0.0001	0.0005
MOTOR	X	6	5.6451	5.6455	5.6519	0.0001	0.0005
EHRSHOT	X	117	5.4965	5.5514	5.5680	0.0005	0.0005
EHRSHOT	X	17	5.5675	5.5857	5.6044	0.0005	0.0005
EHRSHOT	X	6	6.0506	6.0767	6.0839	0.0005	0.0005
TRADE	X	117	5.2599	5.2480	5.2807	0.0005	0.0005
TRADE	X	17	5.4538	5.4529	5.4839	0.0005	0.0005
TRADE	X	6	6.0481	6.0643	6.0702	0.0005	0.0005
EHRmamba	X	117	5.1366	5.1603	5.2125	0.0005	0.0005
EHRmamba	X	17	5.4389	5.4228	5.4678	0.0005	0.0005
EHRmamba	X	6	5.9261	5.9404	5.9573	0.0005	0.0005
Ours (No value share)	X	117	4.6861	4.6386	4.6273	0.0005	0.0005
Ours (No value share)	X	17	4.9070	4.8621	4.8664	0.0005	0.0005
Ours (No value share)	X	6	5.3675	5.3538	5.4045	0.0005	0.0005
FM4EHR	O	117	6.4288	6.4611	6.4344	0.0005	0.0001
FM4EHR	O	17	6.3888	6.4391	6.3969	0.0005	0.0005
FM4EHR	O	6	6.3972	6.4394	6.4069	0.0005	0.0005
ETHOS	O	117	4.9710	5.0374	4.9378	0.0005	0.0001
ETHOS	O	17	5.2479	5.2044	5.1124	0.0005	0.0001
ETHOS	O	6	5.5724	5.5714	5.5213	0.0005	0.0001
STraTS	O	117	5.7857	5.8492	5.8505	0.0001	0.0005
STraTS	O	17	5.8123	5.8335	5.8446	0.0001	0.0005
STraTS	O	6	6.0711	6.0760	6.0834	0.0001	0.0005
Ours (Value share)	O	117	4.8786	4.8954	5.0276	0.0005	0.0005
Ours (Value share)	O	17	5.0430	5.0862	5.1887	0.0005	0.0001
Ours (Value share)	O	6	5.5613	5.6554	5.7523	0.0005	0.0005

was selected from 5×10^{-4} , 1×10^{-4} , 5×10^{-5} , 1×10^{-5} . The batch size was fixed 100. Training was performed for 5 epochs with a 50-step warm-up followed by cosine annealing that decayed the learning rate to 1/100 of its initial value.

Each task-specific prediction head consisted of a two-layer MLP. To account for label imbalance, task losses were weighted according to label frequencies in the training dataset. For efficiency, downstream tasks were conducted in a multi-task setting where all seven tasks were jointly optimized; the final objective was defined as the average of the task-specific losses. Since probing does not allow training of new tokens, we instead introduced a `<Birth>` token at each timestep, serving the same role as the `<SOS>` token, and used its representation for task prediction. For baseline models, we preserved their original binning and embedding procedures, while unifying the learning objective to the downstream tasks. All downstream tasks were conducted on a single NVIDIA A40 GPU.

F. Results

F.1. Downstream Task results

Tables 14, 15, and 16 report the results of all downstream tasks. Table 14 presents the loss, number of features used, and selected pretraining/downstream-task learning rate for each experiment, while the remaining results are shown in Tables 15 and 16.

Table 15. Performance Results (Part 1); I., and P. denote In-Hospital-Mortality, and Phenotyping respectively. R., and P. denote AUROC and AUPRC respectively.

Methods	I.-R.	I.-P.	P.-R-@0	P.-P0	P.-R.1	P.-R.2	P.-P2	P.-R.3	P.-R.4	P.-P4	P.-R.5	P.-P.5	P.-R.6	P.-R.7	P.-P7	P.-R.8	P.-P8	P.-R.9	P.-P.9	P.-R.10	P.-R.11	P.-P11	P.-R.12	P.-P12	P.-R.13	P.-P13	P.-R.14	P.-P14	P.-R.15	P.-P15	P.-R.16	P.-P16					
HEART	X	0.8379	0.4453	0.7584	0.3648	0.7973	0.2344	0.7145	0.2320	0.6513	0.4095	0.7320	0.2590	0.6748	0.2149	0.6174	0.2753	0.5177	0.7211	0.4108	0.7147	0.3330	0.6545	0.4657	0.6064	0.5710	0.6996	0.4352	0.6489	0.1730	0.7242	0.2543					
HEART	X	0.8198	0.3859	0.7779	0.3948	0.7932	0.2383	0.6423	0.1757	0.6187	0.4145	0.6870	0.2286	0.6679	0.2005	0.6167	0.2811	0.6886	0.1545	0.6910	0.4164	0.7309	0.5993	0.8415	0.4566	0.7287	0.3826	0.4917	0.6679	0.4401	0.6260	0.1110	0.6856	0.2256			
HEART	X	0.7391	0.2829	0.6813	0.3525	0.8732	0.1459	0.6335	0.1574	0.6168	0.4019	0.6931	0.1798	0.6707	0.2113	0.5934	0.2550	0.6925	0.1342	0.6895	0.4207	0.7694	0.4991	0.6117	0.1442	0.5801	0.2405	0.7233	0.5048	0.7077	0.5125	0.6302	0.3732	0.5593	0.0971	0.6131	0.1899
MOTOR	X	0.8724	0.5465	0.8392	0.3832	0.8732	0.4259	0.7342	0.2799	0.7162	0.5116	0.8596	0.5353	0.7506	0.3126	0.6691	0.3434	0.7544	0.1765	0.7920	0.5770	0.7938	0.6587	0.7566	0.2974	0.6490	0.2779	0.7233	0.5048	0.7100	0.6214	0.7354	0.4857	0.7443	0.1887	0.8503	0.4827
MOTOR	X	0.8406	0.4612	0.7440	0.4240	0.8743	0.4176	0.6969	0.2043	0.6940	0.4910	0.7501	0.3115	0.7293	0.2733	0.6648	0.3287	0.7426	0.1707	0.7492	0.4966	0.7812	0.6392	0.7575	0.3195	0.6580	0.2888	0.7953	0.4761	0.6762	0.7788	0.6899	0.4337	0.6892	0.1472	0.7348	0.2858
ETHSHOT	X	0.8107	0.4895	0.6638	0.3446	0.6405	0.1153	0.5722	0.1332	0.5895	0.3900	0.6761	0.2300	0.5557	0.1427	0.6075	0.2900	0.5849	0.0890	0.6229	0.3531	0.6390	0.4345	0.9844	0.1240	0.5581	0.1918	0.5834	0.3701	0.5559	0.4997	0.6108	0.3429	0.5610	0.0980	0.6805	0.2363
ETHSHOT	X	0.8107	0.4240	0.6034	0.2693	0.6670	0.1243	0.5573	0.1329	0.5695	0.3884	0.6858	0.1520	0.5719	0.1624	0.5989	0.2943	0.5632	0.0831	0.6007	0.3464	0.5310	0.4019	0.5972	0.1553	0.5612	0.2237	0.5959	0.3744	0.5356	0.4589	0.5791	0.3163	0.5453	0.0928	0.6021	0.1647
ETHSHOT	X	0.6960	0.3102	0.6083	0.2771	0.5904	0.0855	0.5815	0.1507	0.5833	0.4038	0.5366	0.1450	0.5690	0.1547	0.5965	0.2959	0.5673	0.0893	0.6185	0.3683	0.5533	0.3895	0.4837	0.0910	0.5534	0.2231	0.5261	0.3056	0.5252	0.4465	0.5095	0.3099	0.5253	0.0884	0.5369	0.1438
TRADE	X	0.8276	0.4406	0.8314	0.5590	0.8106	0.2813	0.7246	0.2258	0.6977	0.4851	0.8236	0.4264	0.7015	0.2542	0.6618	0.3333	0.7102	0.1543	0.7762	0.5430	0.7747	0.6194	0.7902	0.2961	0.6473	0.2784	0.6941	0.4793	0.6206	0.5323	0.7279	0.4886	0.7143	0.1669	0.8214	0.4197
TRADE	X	0.8107	0.4895	0.6638	0.3446	0.6405	0.1153	0.5722	0.1332	0.5895	0.3900	0.6761	0.2300	0.5557	0.1427	0.6075	0.2900	0.5849	0.0890	0.6229	0.3531	0.6390	0.4345	0.9844	0.1240	0.5581	0.1918	0.5834	0.3701	0.5559	0.4997	0.6108	0.3429	0.5610	0.0980	0.6805	0.2363
TRADE	X	0.6726	0.2184	0.6533	0.3408	0.6695	0.1498	0.6070	0.1802	0.6032	0.4188	0.6141	0.1826	0.6153	0.2857	0.6280	0.1152	0.6652	0.1480	0.6647	0.4574	0.9814	0.1332	0.5835	0.2301	0.5841	0.3660	0.6041	0.5224	0.6034	0.3469	0.5338	0.1102	0.5995	0.1757		
ETHRumba	X	0.8677	0.5865	0.7765	0.5018	0.7149	0.1678	0.6756	0.3126	0.6447	0.4364	0.7657	0.2894	0.6304	0.1736	0.5904	0.2567	0.6865	0.1406	0.7236	0.4395	0.7457	0.6117	0.6567	0.1330	0.6077	0.2467	0.6345	0.4229	0.5919	0.4943	0.6973	0.4410	0.6709	0.1419	0.7462	0.2658
ETHRumba	X	0.8125	0.4016	0.6265	0.3172	0.7663	0.1601	0.6672	0.1769	0.6328	0.4456	0.6588	0.2140	0.5783	0.1578	0.6149	0.2856	0.6661	0.1245	0.6319	0.3601	0.6848	0.5227	0.5842	0.1347	0.5962	0.2386	0.6663	0.4479	0.5554	0.6484	0.6194	0.3713	0.5060	0.0874	0.6245	0.1906
ETHRumba	X	0.7048	0.3109	0.6112	0.2996	0.6305	0.1175	0.5695	0.1370	0.5785	0.3952	0.5616	0.1582	0.5843	0.1593	0.5997	0.2835	0.6672	0.1240	0.6283	0.3596	0.7142	0.5345	0.5863	0.1259	0.5662	0.3363	0.4935	0.4181	0.5733	0.3232	0.5272	0.0907	0.5794	0.1624		
Ours (No value share)	X	0.8891	0.6869	0.7863	0.6506	0.8992	0.4789	0.8412	0.5324	0.7492	0.5938	0.9041	0.6281	0.7616	0.3442	0.6994	0.3482	0.7902	0.2504	0.8273	0.6372	0.8321	0.7077	0.8711	0.4865	0.7581	0.4072	0.7426	0.5281	0.7292	0.6327	0.7648	0.5438	0.8219	0.3900	0.8954	0.5912
Ours (No value share)	X	0.8694	0.5446	0.7552	0.4393	0.8902	0.4623	0.7563	0.2751	0.7228	0.5311	0.7828	0.3644	0.7305	0.2812	0.6859	0.3440	0.7652	0.2255	0.7689	0.5301	0.8057	0.6630	0.8538	0.4809	0.7600	0.4059	0.7271	0.5027	0.6883	0.5842	0.7133	0.4612	0.6922	0.1618	0.7618	0.3246
Ours (No value share)	X	0.8012	0.4016	0.7237	0.4052	0.7655	0.2476	0.7265	0.2807	0.7200	0.5348	0.7229	0.2800	0.7274	0.2751	0.6862	0.3181	0.7596	0.2041	0.7515	0.5308	0.7840	0.6057	0.8087	0.2653	0.6203	0.3499	0.6956	0.4573	0.6808	0.5771	0.6749	0.4409	0.6170	0.1207	0.7228	0.2289
FMAHER	O	0.6172	0.1770	0.4513	0.1876	0.5803	0.0886	0.4661	0.0980	0.4534	0.2972	0.5673	0.1619	0.5155	0.1266	0.5571	0.2488	0.1866	0.0567	0.5245	0.2789	0.5075	0.3785	0.5384	0.1065	0.5043	0.1923	0.5176	0.2966	0.5315	0.4459	0.5974	0.3251	0.4676	0.0735	0.5518	0.1572
FMAHER	O	0.5773	0.1519	0.5000	0.2121	0.5000	0.0661	0.5000	0.1076	0.5000	0.3231	0.8070	0.1524	0.5800	0.1257	0.5000	0.2130	0.5000	0.0704	0.5000	0.2862	0.5000	0.3305	0.5000	0.0845	0.5800	0.1920	0.5000	0.2889	0.5000	0.1227	0.5000	0.2686	0.5000	0.0789	0.5000	0.1305
FMAHER	O	0.4965	0.1611	0.4500	0.1800	0.4800	0.0600	0.4500	0.0700	0.4500	0.2600	0.7000	0.1200	0.4500	0.0500	0.4500	0.1300	0.4500	0.0400	0.4500	0.2500	0.4500	0.3500	0.4500	0.0700	0.4500	0.1800	0.4500	0.2400	0.4500	0.1000	0.4500	0.2000	0.4500	0.0300	0.4500	0.0500
ETHOS	O	0.8173	0.4162	0.7222	0.3959	0.8489	0.3805	0.7014	0.2017	0.6914	0.4886	0.7233	0.2830	0.6767	0.2344	0.6535	0.3293	0.7383	0.1721	0.7432	0.4919	0.7721	0.6178	0.7536	0.3158	0.6726	0.3013	0.7011	0.4855	0.6587	0.5597	0.6732	0.4109	0.6596	0.1173	0.6967	0.2395
ETHOS	O	0.7558	0.3269	0.6759	0.3660	0.7145	0.1814	0.6554	0.1679	0.6861	0.4676	0.6209	0.1947	0.6706	0.2016	0.6385	0.3195	0.7257	0.1507	0.7105	0.4579	0.7477	0.5695	0.5869	0.1401	0.5965	0.2280	0.6741	0.4415	0.6501	0.5504	0.6548	0.3605	0.8881	0.1015	0.6127	0.1779
STATS	O	0.7592	0.1311	0.6583	0.3273	0.7350	0.1876	0.6662	0.1775	0.6936	0.4720	0.8849	0.2426	0.6735	0.1852	0.5766	0.2465	0.4506	0.1617	0.7300	0.4761	0.7466	0.5275	0.6101	0.1570	0.6048	0.2384	0.6422	0.3686	0.6135	0.9866	0.3993	0.3262	0.5994	0.1081	0.6735	0.2206
STATS	O	0.6993	0.2192	0.6472	0.3141	0.6621	0.1421	0.6552	0.1708	0.6863	0.4632	0.6695	0.2322	0.6747	0.1803	0.5862	0.2477	0.7445	0.1654	0.7213	0.4679	0.7370	0.5289	0.6942	0.1459	0.5940	0.2299	0.6174	0.3715	0.6217	0.5964	0.6084	0.3324	0.5547	0.0969	0.6636	0.2067
Ours (Value share)	O	0.6760	0.5586	0.8540	0.5745	0.8942	0.4188	0.8394	0.5054	0.7162	0.5124	0.8512	0.4999	0.7152	0.2714	0.6800	0.3453	0.5723	0.1835	0.8091	0.6981	0.7379	0.7040	0.8403	0.4538	0.6886	0.3019	0.7278	0.5097	0.6678	0.5782	0.7534	0.4967	0.7677	0.1027	0.8345	0.4477
Ours (Value share)	O	0.8649	0.5191	0.8297	0.5470	0.8603	0.3063	0.7747	0.4233	0.6637	0.4540	0.7977	0.3551	0.6362	0.1834	0.6466	0.2999	0.7072	0.1478	0.7331	0.4609	0.7776	0.6534	0.7502	0.2389	0.6360	0.2687	0.6762	0.4728	0.6171	0.5319	0.7419	0.4725	0.7141	0.1599	0.7680	0.3210
Ours (Value share)	O	0.7751	0.3660	0.7033	0.3846	0.7115	0.1505	0.6880	0.1810	0.6705	0.5073	0.7126	0.2656	0.7096	0.2347	0.6440	0.3138	0.7348	0.1635	0.7359	0.4696	0.7731	0.6695	0.6084	0.1393	0.6039	0.2362	0.7043	0.4712	0.6479	0.5555	0.6754	0.4082	0.5690	0.0979	0.6799	0.2386

Table 16. Performance Results (Part 2); P, L, H., and V. denote Phenotyping, Length-of-Stay, Oliguria/Anuria, and Vasopressor respectively. R., and P. denote AUROC and AUPRC respectively.

Method	Value sharing	P-R-17	P-R-18	P-R-19	P-R-20	P-R-21	P-R-22	P-R-23	P-R-24	P-R-25	P-R-26	P-R-27	P-R-28	P-R-29	P-R-30	H-R-1	H-R-10	H-R-11	H-macro R.	H-micro R.	V.-R.	V.-P.																
HEART	X	0.7663	0.2522	0.6253	0.0866	0.6792	0.0899	0.6448	0.1344	0.6310	0.1774	0.5902	0.1152	0.6797	0.2104	0.6436	0.1797	0.6245	0.7008	0.6552	0.2559	0.6649	0.2723	0.6749	0.1229	0.6304	0.1821	0.1417	2.1418	0.6676	0.4354	0.7383	0.7029	0.7014	0.8652	0.3629		
HEART	X	0.6454	0.1041	0.6448	0.0877	0.6494	0.0839	0.6339	0.1370	0.6249	0.1755	0.6024	0.1101	0.6410	0.2012	0.6536	0.1834	0.6097	0.0665	0.6326	0.2499	0.6468	0.2610	0.6357	0.1210	0.6258	0.1789	0.1361	2.1316	0.6274	0.3674	0.6834	0.2519	0.6454	0.6572	0.8534	0.3434	
HEART	X	0.6442	0.1424	0.6447	0.0877	0.6494	0.0839	0.6339	0.1370	0.6249	0.1755	0.6024	0.1101	0.6410	0.2012	0.6536	0.1834	0.6097	0.0665	0.6326	0.2499	0.6468	0.2610	0.6357	0.1210	0.6258	0.1789	0.1361	2.1316	0.6274	0.3674	0.6834	0.2519	0.6454	0.6572	0.8534	0.3434	
HEART	X	0.7663	0.2522	0.6253	0.0866	0.6792	0.0899	0.6448	0.1344	0.6310	0.1774	0.5902	0.1152	0.6797	0.2104	0.6436	0.1797	0.6245	0.7008	0.6552	0.2559	0.6649	0.2723	0.6749	0.1229	0.6304	0.1821	0.1417	2.1418	0.6676	0.4354	0.7383	0.7029	0.7014	0.8652	0.3629		
MOTOR	X	0.6971	0.2001	0.6092	0.1128	0.6720	0.0998	0.6552	0.1573	0.6529	0.2006	0.6192	0.1331	0.6725	0.2258	0.6506	0.1919	0.6364	0.0751	0.6684	0.2717	0.6779	0.2828	0.6677	0.1374	0.6628	0.2019	0.1585	2.0095	0.6574	0.4156	0.7006	0.2952	0.6790	0.6869	0.8708	0.3398	
MOTOR	X	0.6309	0.1669	0.5718	0.0621	0.6091	0.0669	0.5807	0.1135	0.6302	0.1995	0.7018	0.3212	0.6226	0.2025	0.6799	0.1635	0.5951	0.5987	0.8556	0.1607	0.8397	0.1550	0.2972	0.1324	0.2972	0.1324	0.1084	0.5803	0.3459	0.6037	0.2100	0.5965	0.6322	0.8404	0.3190		
EHRSPLIT	X	0.5707	0.1169	0.5718	0.0609	0.6049	0.0669	0.5807	0.1135	0.6302	0.1995	0.7018	0.3212	0.6226	0.2025	0.6799	0.1635	0.5951	0.5987	0.8556	0.1607	0.8397	0.1550	0.2972	0.1324	0.2972	0.1324	0.1084	0.5803	0.3459	0.6037	0.2100	0.5965	0.6322	0.8404	0.3190		
EHRSPLIT	X	0.8352	0.4457	0.6887	0.1122	0.7183	0.1203	0.6831	0.1694	0.6672	0.2128	0.6356	0.1429	0.7131	0.2406	0.6705	0.2017	0.6553	0.0797	0.6948	0.2877	0.7151	0.3006	0.6757	0.1399	0.6729	0.2031	0.1673	2.0490	0.7118	0.5069	0.7860	0.4370	0.7552	0.7495	0.8997	0.4423	
TRADE	X	0.8038	0.3821	0.6877	0.1025	0.6746	0.1102	0.6481	0.1592	0.6403	0.2032	0.6117	0.1362	0.6797	0.2267	0.6520	0.1845	0.5975	0.0732	0.6292	0.2626	0.6417	0.2728	0.6014	0.1260	0.6464	0.1260	0.6464	0.1260	2.1287	0.5922	0.4099	0.6872	0.3256	0.6249	0.6229	0.8919	0.2645
TRADE	X	0.7261	0.3440	0.5929	0.0942	0.6307	0.1100	0.6469	0.1465	0.5909	0.1890	0.5725	0.1282	0.6432	0.2127	0.6131	0.1803	0.5975	0.0732	0.6292	0.2626	0.6417	0.2728	0.6014	0.1260	0.6464	0.1260	0.6464	0.1260	2.1287	0.5922	0.4099	0.6872	0.3256	0.6249	0.6229	0.8919	0.2645
EHREnrich	X	0.8724	0.5777	0.6428	0.1083	0.6958	0.1169	0.6566	0.1654	0.6406	0.2096	0.6117	0.1397	0.6820	0.2389	0.6468	0.1952	0.6324	0.0763	0.6766	0.2843	0.6845	0.2975	0.6576	0.1261	0.6550	0.1921	0.1660	2.0724	0.7051	0.5030	0.7821	0.4299	0.7501	0.7437	0.8932	0.4401	
EHREnrich	X	0.8109	0.3917	0.6101	0.0997	0.6577	0.1074	0.6279	0.1557	0.6151	0.1981	0.5845	0.1330	0.6589	0.2241	0.6192	0.1892	0.6045	0.0763	0.6458	0.2721	0.6882	0.2851	0.6212	0.1320	0.6224	0.1948	0.1477	2.1357	0.6553	0.4540	0.7234	0.3749	0.6942	0.6895	0.8398	0.3532	
EHREnrich	X	0.7194	0.2493	0.5786	0.0912	0.6105	0.0965	0.5891	0.1444	0.5854	0.1865	0.5869	0.1247	0.6197	0.2047	0.5954	0.1777	0.5806	0.0732	0.6126	0.2621	0.6218	0.2695	0.5827	0.1366	0.5855	0.1879	0.1203	2.2146	0.5754	0.3862	0.6410	0.3109	0.6110	0.6093	0.7571	0.2468	
One (No value share)	X	0.8911	0.6022	0.7458	0.1487	0.8153	0.1835	0.7752	0.2361	0.7685	0.2828	0.7521	0.1923	0.8102	0.3317	0.7718	0.2530	0.7511	0.1003	0.8027	0.3739	0.8272	0.3951	0.7804	0.1869	0.7756	0.2387	0.1974	1.9608	0.7739	0.5744	0.8490	0.4807	0.8126	0.8084	0.9186	0.4978	
One (No value share)	X	0.8865	0.5293	0.6987	0.1134	0.7208	0.1537	0.6831	0.1994	0.6760	0.2403	0.6996	0.1627	0.7129	0.2625	0.6797	0.2193	0.6624	0.0893	0.7014	0.3274	0.7337	0.3459	0.6865	0.1543	0.6810	0.2296	0.1628	2.0943	0.6912	0.4935	0.7557	0.3742	0.7210	0.7167	0.8469	0.3592	
One (No value share)	X	0.8110	0.4084	0.6523	0.1190	0.7208	0.1537	0.6831	0.1994	0.6760	0.2403	0.6996	0.1627	0.7129	0.2625	0.6797	0.2193	0.6624	0.0893	0.7014	0.3274	0.7337	0.3459	0.6865	0.1543	0.6810	0.2296	0.1628	2.0943	0.6912	0.4935	0.7557	0.3742	0.7210	0.7167	0.8469	0.3592	
FM4EHR	O	0.6221	0.1842	0.5900	0.0731	0.6190	0.0831	0.5900	0.1348	0.5900	0.1656	0.5900	0.1097	0.5900	0.1816	0.5900	0.1332	0.5900	0.0601	0.5900	0.3308	0.5900	0.2415	0.5900	0.1029	0.5900	0.1607	0.1183	2.2017	0.5900	0.3136	0.5900	0.2102	0.5900	0.5900	0.5900	0.1451	
FM4EHR	O	0.4928	0.1147	0.5000	0.0731	0.5000	0.0831	0.5000	0.1248	0.5000	0.1656	0.5000	0.1097	0.5000	0.1816	0.5000	0.1332	0.5000	0.0601	0.5000	0.3308	0.5000	0.2415	0.5000	0.1029	0.5000	0.1607	0.1119	2.2917	0.5900	0.3136	0.5900	0.2102	0.5900	0.5900	0.5900	0.1451	
FM4EHR	O	0.8272	0.5290	0.6809	0.1176	0.7441	0.1251	0.7066	0.1746	0.6898	0.2191	0.6604	0.1463	0.7369	0.2517	0.6960	0.2123	0.6798	0.0839	0.7197	0.2978	0.7385	0.3117	0.6951	0.1451	0.6920	0.2094	0.1708	2.0334	0.7107	0.5089	0.7850	0.4407	0.7552	0.7521	0.8979	0.4419	
ETROS	O	0.8421	0.4441	0.6422	0.1079	0.7666	0.1160	0.6669	0.1629	0.6529	0.2055	0.6248	0.1376	0.6707	0.2257	0.6560	0.1923	0.6364	0.0731	0.6468	0.2661	0.6401	0.2991	0.6245	0.1302	0.6288	0.1892	0.1254	2.1602	0.6166	0.4108	0.6812	0.3235	0.6439	0.6420	0.7942	0.2749	
ETROS	O	0.7559	0.3306	0.6120	0.0998	0.6482	0.1087	0.6289	0.1531	0.6153	0.1945	0.5800	0.1316	0.6698	0.2209	0.6333	0.1660	0.6184	0.0731	0.6468	0.2661	0.6401	0.2991	0.6245	0.1302	0.6288	0.1892	0.1254	2.1602	0.6166	0.4108	0.6812	0.3235	0.6439	0.6420	0.7942	0.2749	
STFATS	O	0.7010	0.3076	0.6120	0.1003	0.6732	0.1098	0.6310	0.1574	0.6176	0.1952	0.5910	0.1311	0.6694	0.2224	0.6349	0.1872	0.6204	0.0734	0.6510	0.2676	0.6658	0.2817	0.6291	0.1302	0.6298	0.1892	0.1280	2.1628	0.6131	0.4070	0.6810	0.3245	0.6426	0.6402	0.7942	0.2749	
STFATS	O	0.7088	0.2459	0.5859	0.0921	0.6408	0.1015	0.6031	0.1479	0.5964	0.1851	0.5724	0.1249	0.6489	0.2130	0.6132	0.1810	0.5989	0.0716	0.6297	0.2557	0.6613	0.2693	0.6055	0.1254	0.6046	0.1821	0.1097	2.2125	0.5820	0.3752	0.6410	0.2940	0.6074	0.6053	0.7901	0.2325	
One (Value share)	O	0.8789	0.5617	0.7057	0.1441	0.7925	0.1759	0.7567	0.2289	0.7430	0.2752	0.7154	0.1870	0.7856	0.3099	0.7504	0.2537	0.7338	0.0979	0.7808	0.3642	0.8049	0.3873	0.7551	0.1768	0.7518	0.2536	0.1890	1.9917	0.7556	0.5560	0.8249	0.4549	0.7903	0.7859	0.9115	0.4762	
One (Value share)	O	0.8540	0.4998	0.6642	0.1269	0.7501	0.1638	0.7186	0.2095	0.6740	0.2619	0.7155	0.1824	0.7696	0.3012	0.7254	0.2544	0.7096	0.0912	0.7427	0.3426	0.8049	0.3873	0.7551	0.1768	0.7518	0.2536	0.1890	1.9917	0.7556	0.5560	0.8249	0.4549	0.7903	0.7859	0.9115	0.4762	
One (Value share)	O	0.7848	0.3864	0.6210	0.1154	0.6996	0.1190	0.6588	0.1901	0.6505	0.2280	0.6244	0.1564	0.6912	0.2619	0.6607	0.2154	0.6456	0.0852	0.6780	0.3128	0.7089	0.3352	0.6597	0.1472	0.6588	0.2278	0.1529	2.1220	0.6631	0.4480	0.7278	0.3598	0.6943	0.6899	0.8520	0.3229	

F.2. Detailed Analysis

To complement the main ablation, we provide additional analyses examining (1) whether our gains arise from model capacity, and (2) the sensitivity of each proposed component to its key hyperparameters. All scores are reported as Total Score ($\downarrow$), with IHM AUROC/AUPRC where relevant.

Model capacity is not the source of our gains. We first verify that our improvements are not simply a matter of parameter count. As shown in Table 17, increasing the baseline backbones from 6 to 8 layers ($\sim 30\%$ more parameters) yields only marginal changes ($< 1\%$), and the enlarged baselines still trail our 6-layer model. This indicates that our performance stems from the proposed architectural methodologies rather than added capacity.

Pathology-Focused Binning. (30-epoch) We analyze the two main hyperparameters of our binning scheme. Table 18 varies the density-weighting exponent λ : any $\lambda > 0$ improves over plain quantile binning ($\lambda = 0$), with the best results around $\lambda = 1$, our default. Table 19 varies the number of bins. Without value sharing, performance peaks at 10 bins and degrades beyond it as per-feature value tokens become sparse; with value sharing, the shared vocabulary tolerates finer granularity, and performance continues to improve up to 50 bins. This confirms that the optimal resolution depends on the tokenization scheme, while our default of 10 bins is a robust choice in the non-shared setting.

Dual-Calendar RoPE. (30-epoch) Finally, Table 20 examines the number of calendrical dimensions, reduced either by sparsifying periods while fixing the endpoints (“Fix both ends”) or by retaining only the high-frequency periods (“Fix high-freqs”). Performance is stable around our default of 40 dimensions but degrades markedly once the calendar dimension drops below 20 under either scheme, underscoring the importance of covering a wide range of clinically meaningful periods.

Table 17. **Baseline capacity analysis.** Increasing baseline depth from 6 to 8 layers gives only minor gains while still trailing our 6-layer model.

(50-epoch)	Best (MOTOR)			Avg. (8 baselines)		
	Total $\downarrow$	IHM ROC	IHM PRC	Total $\downarrow$	IHM ROC	IHM PRC
6-layer	4.889	0.880	0.552	5.416	0.810	0.434
8-layer	4.953	0.878	0.554	5.390	0.817	0.438

Table 18. **Binning: weighting λ .** Weighted binning ($\lambda > 0$) improves over quantile binning, best around $\lambda = 1$.

Binning Analysis	$\lambda = 0$ (Quantile)	0.5	1 (Ours)	1.5	2
Total score ($\downarrow$)	4.827	4.800	4.762	4.800	4.790
IHM AUROC	0.877	0.878	0.881	0.885	0.882
IHM AUPRC	0.580	0.584	0.596	0.587	0.583

Table 19. **Binning: bin count.** With value sharing, performance improves up to 50 bins; without sharing, it degrades beyond 10 bins due to sparsity.

Number of bins	2	5	10 (Default)	20	50
No value share	4.930	4.790	4.762	4.797	4.830
Value share	5.208	5.095	4.985	4.906	4.883

F.3. EHR generation and evaluation

We generated EHRs with a low temperature of 0.2. Below are examples generated from the same initial EHR, shown in order for our model (Figure 7) and the ETHOS model (Figure 8).

Figure 9 shows the prompt we used for LLM-based evaluation.

Table 20. **Dual-Calendar RoPE: calendrical components.** Performance degrades once the calendar dimension drops below 20, under both reduction schemes.

PE analysis	Calendar dim					
	40 (Ours)	30	20	10	6	0 (RoPE)
Fix both ends	4.762	4.755	4.789	4.841	4.842	4.861
Fix high-freqs	4.762	4.759	4.767	4.802	4.855	4.861

G. LLM usage clarification

In addition to the uses of LLMs described in the main text, we employed them for summarizing content, translation, grammar correction, and sentence refinement during the writing of the manuscript. In the early stages of the study, we used LLMs to search for related work, and the retrieved papers were then read and verified by the researchers.

Ours (share)

Birth : 1845. 05. 11	hemoglobin : 31.0	- GCS-V : 5	- MBP : 72
Sex: Female	- Mean corpuscular	- HR : 117	- O2 saturation : 79
Ethnicity: WHITE	hemoglobin concentra-	- MBP : 73	- RR : 19
Age: 300	tion : 34.4	- O2 saturation : 92	- SBP : 116
	- Mean corpuscular volume	- RR : 19	2145-05-12 06:00:00
2145-05-11 17:12:55	: 90.0	- SBP : 115	- DBP : 47
- ICU transfer	- Platelets : 200.0	- Temperature : 36.9	- HR : 114
2145-05-11 17:30:00	- Prothrombin time : 13.3	2145-05-12 00:00:00	- MBP : 70
- DBP : 62	- Partial thromboplastin	- SBP : 116	- O2 saturation : 92
- SBP : 103	time : 32.4	- RR : 19	- RR : 20
- HR : 123	- Red Cell Distribution	- O2 saturation : 91	- SBP : 107
- MBP : 83	Width : 13.3	- HR : 118	2145-05-12 07:00:00
- RR : 4	- Red blood cell count :	- DBP : 53	- HR : 111
2145-05-11 18:00:00	3.74	- MBP : 74	- DBP : 49
- DBP : 51	- White blood cell count :	2145-05-12 01:00:00	- GCS-E : 4
- HR : 110	12.20	- DBP : 50	- GCS-M : 6
- MBP : 73	- RR : 18	- HR : 116	- GCS : 14
- RR : 17	2145-05-11 19:00:00	- MBP : 76	- GCS-V : 5
- SBP : 111	- DBP : 56	- O2 saturation : 93	- MBP : 73
—# Gen Start #—	- HR : 119	- RR : 18	- O2 saturation : 93
- O2 saturation : 90	- MBP : 73	- SBP : 113	- RR : 19
- Weight : 54.2	- O2 saturation : 92	2145-05-12 02:00:00	- SBP : 114
- Temperature : 36.7	- RR : 19	- HR : 111	- Temperature : 36.6
- GCS : 15	- SBP : 111	- DBP : 49	2145-05-12 08:00:00
- GCS-M : 6	2145-05-11 20:00:00	- MBP : 74	- DBP : 52
- GCS-E : 4	- DBP : 50	- RR : 20	- HR : 112
- GCS-V : 5	- MBP : 76	- SBP : 116	- MBP : 71
- O2 saturation : 92	- HR : 110	- O2 saturation : 92	- O2 saturation : 89
- HR : 117	- RR : 19	2145-05-12 03:00:00	- RR : 19
- SBP : 108	- SBP : 111	- DBP : 52	- SBP : 114
- Glucose : 261	- O2 saturation : 90	- GCS-E : 4	2145-05-12 09:00:00
- MBP : 76	2145-05-11 21:00:00	- GCS-M : 6	- DBP : 52
- DBP : 50	- DBP : 51	- GCS : 15	- HR : 116
- Anion gap : 15.0	- HR : 113	- GCS-V : 5	- MBP : 76
- Bicarbonate : 24.0	- MBP : 71	- HR : 119	- O2 saturation : 88
- Calcium Total : 8.6	- O2 saturation : 93	- MBP : 72	- RR : 20
- Chloride : 105.0	- RR : 19	- O2 saturation : 91	- SBP : 110
- Creatinine : 0.9	- SBP : 115	- RR : 20	2145-05-12 10:00:00
- Magnesium : 1.8	2145-05-11 22:00:00	- SBP : 109	- DBP : 53
- Phosphate : 2.7	- DBP : 50	- Temperature : 36.9	- HR : 113
- Potassium : 4.0	- HR : 114	2145-05-12 04:00:00	- MBP : 75
- Sodium : 142.0	- MBP : 74	- DBP : 50	- O2 saturation : 92
- Blood urea nitrogen :	- O2 saturation : 89	- HR : 117	- RR : 20
12.0	- RR : 19	- MBP : 73	- SBP : 111
- Hematocrit : 33.0	- SBP : 116	- RR : 18	2145-05-12 11:00:00
- Hemoglobin chemistry :	2145-05-11 23:00:00	- SBP : 113	- DBP : 52
11.3	- DBP : 51	- O2 saturation : 85	- GCS-E : 4
- Prothrombin time INR :	- GCS-E : 4	2145-05-12 05:00:00	- GCS-M : 6
1.10	- GCS-M : 6	- DBP : 49	...
- Mean corpuscular	- GCS : 15	- HR : 116	

Figure 7. Temporal EHR history generated by our model (used value share version for fair comparison). The data before this “—# Gen Start #—” marker is given, and the data after it is generated.

ETHOS

Birth : 1845. 05. 11	- DBP : 42	- O2 saturation : 85	2145-05-12 05:05:00
Sex: Female	- MBP : 60	- HR : 119	- HR : 115
Ethnicity: WHITE	2145-05-11 20:45:00	- DBP : 43	- O2 saturation : 93
Age: 300	- RR : 30	- MBP : 60	- RR : 29
	- SBP : 84	2145-05-12 00:30:00	2145-05-12 05:25:00
2145-05-11 17:12:55	- O2 saturation : 93	- RR : 32	- DBP : 31
- ICU transfer	- HR : 115	- HR : 114	- MBP : 59
2145-05-11 17:30:00	- DBP : 44	- O2 saturation : 93	- SBP : 94
- DBP : 62	- MBP : 49	- SBP : 68	2145-05-12 06:25:00
- SBP : 103	2145-05-11 21:25:00	- MBP : 43	- RR : 33
- HR : 123	- HR : 122	- DBP : 43	- O2 saturation : 92
- MBP : 83	- MBP : 58	2145-05-12 01:00:00	- HR : 120
- RR : 4	- O2 saturation : 93	- DBP : 42	2145-05-12 06:45:00
2145-05-11 18:00:00	- RR : 29	- HR : 117	- DBP : 27
- DBP : 51	- SBP : 88	- MBP : 53	- MBP : 55
- HR : 110	- DBP : 40	- O2 saturation : 92	- SBP : 83
- MBP : 73	2145-05-11 22:05:00	- RR : 31	2145-05-12 07:35:00
- RR : 17	- SBP : 81	- SBP : 92	- HR : 119
- SBP : 111	- RR : 29	2145-05-12 01:50:00	- O2 saturation : 91
—# Gen Start #—	- O2 saturation : 87	- HR : 112	- RR : 29
- O2 saturation : 97	- HR : 134	- O2 saturation : 90	2145-05-12 08:05:00
2145-05-11 18:30:00	- DBP : 43	- RR : 33	- DBP : 42
- DBP : 51	- MBP : 59	2145-05-12 02:40:00	- MBP : 42
- HR : 124	2145-05-11 23:00:00	- DBP : 39	- SBP : 86
- MBP : 67	- SBP : 88	- MBP : 48	2145-05-12 09:05:00
- O2 saturation : 97	- RR : 36	- SBP : 91	- HR : 116
- RR : 32	- O2 saturation : 91	2145-05-12 03:15:00	- MBP : 50
- SBP : 89	- HR : 115	- DBP : 43	- O2 saturation : 92
2145-05-11 19:10:00	- DBP : 22	- HR : 120	- RR : 37
- SBP : 80	- MBP : 58	- MBP : 43	- SBP : 81
- RR : 35	2145-05-11 23:30:00	- O2 saturation : 91	- DBP : 43
- O2 saturation : 96	- SBP : 84	- RR : 38	2145-05-12 10:00:00
- HR : 119	- RR : 32	- SBP : 94	- HR : 132
- DBP : 42	- O2 saturation : 92	2145-05-12 04:05:00	- O2 saturation : 92
- MBP : 59	- HR : 137	- RR : 29	- RR : 33
2145-05-11 19:45:00	- DBP : 38	- O2 saturation : 91	2145-05-12 10:15:00
- DBP : 44	- MBP : 49	- HR : 116	- DBP : 35
- HR : 129	2145-05-11 23:45:00	- MBP : 51	- MBP : 55
- MBP : 60	- DBP : 39	- DBP : 38	- SBP : 94
- O2 saturation : 90	- HR : 115	- SBP : 88	2145-05-12 11:10:00
- RR : 30	- MBP : 52	2145-05-12 04:20:00	- DBP : 43
- SBP : 94	- O2 saturation : 91	- RR : 30	- HR : 136
2145-05-11 20:15:00	- RR : 34	- O2 saturation : 87	- MBP : 34
- RR : 32	- SBP : 86	- HR : 123	...
- SBP : 78	2145-05-12 00:15:00	- DBP : 42	
- O2 saturation : 93	- SBP : 88	- MBP : 54	
- HR : 140	- RR : 36	- SBP : 89	

Figure 8. Temporal EHR history generated by ETHOS. The data before this “—# Gen Start #—” marker is given, and the data after it is generated.

You are a physician with extensive ICU experience and an AI researcher familiar with text generation models, such as LLMs.

In this survey, you will compare the quality of EHR texts generated by two different models from the same initial patient history.

The quality of an EHR depends on whether the right clinical events occur at the right times. Please consider both the timing of events and the appropriateness of the events themselves.

First, you will see a few sample ICU EHR texts. Then, for each pair of generated candidates (A and B), you will be asked to decide which one appears more realistic.

<Sample EHR texts>

1. ## Sample 1 ##
2. ## Sample 2 ##
3. ## Sample 3 ##

<end of EHR samples>

<Evaluation candidate A>

ETHOS generated Sample (Random order; Ours can be candidate A)

<Evaluation candidate B>

Ours generated Sample (Random order; ETHOS can be candidate B)

<Compare two candidates>

Figure 9. LLM input prompt for generated EHR evaluation. We compared the generative performance of our model and ETHOS on LLMs with this prompt.